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(54) **CONVERTIBLE PET STROLLER AND CARRIER**

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(51) **Int. Cl.**
A01K 1/02 (2006.01)

(52) **U.S. Cl.**
CPC **A01K 1/0272** (2013.01); **A01K 1/0245** (2013.01)

(58) **Field of Classification Search**

CPC A47D 13/025; B62B 7/14; B62B 7/142; B62B 7/145; B62B 9/102; A01K 1/0272; A01K 1/0245

See application file for complete search history.

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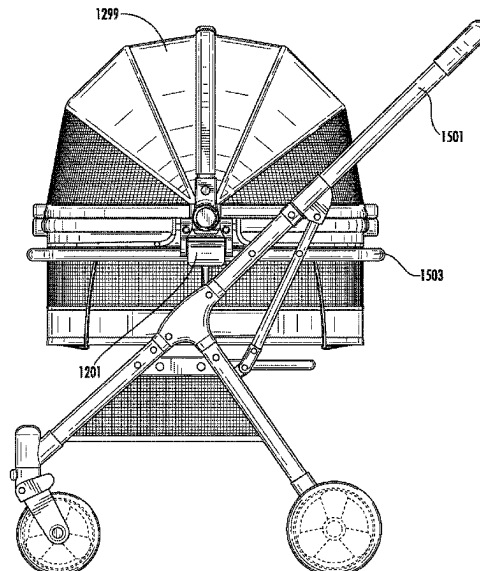
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(57) **ABSTRACT**

An assembly comprising a pet stroller with a removably attached pet carrier. The pet carrier has front and rear access to the enclosed space and is useable separately from the stroller.

15 Claims, 20 Drawing Sheets



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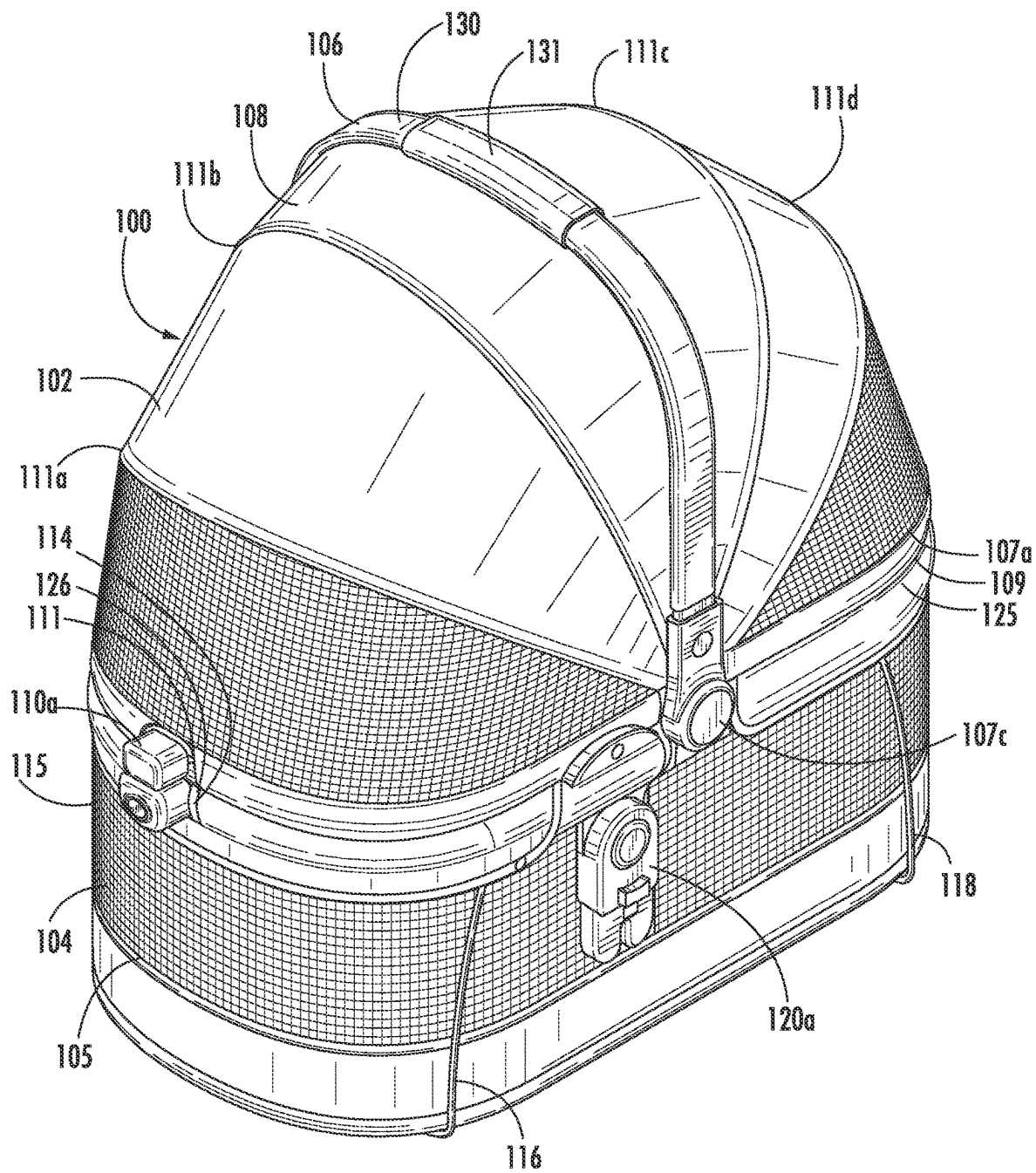


FIG. 1

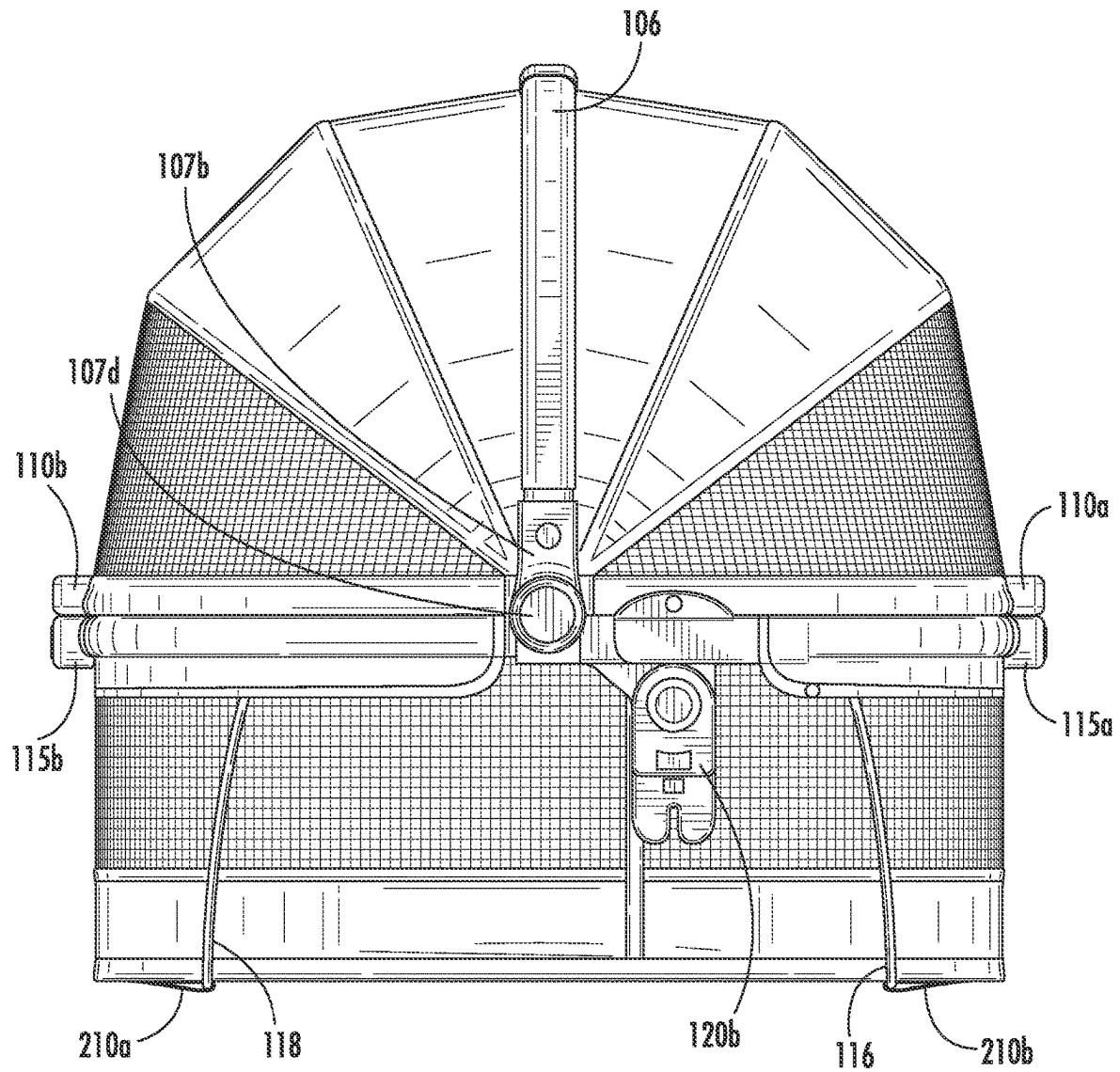


FIG. 2

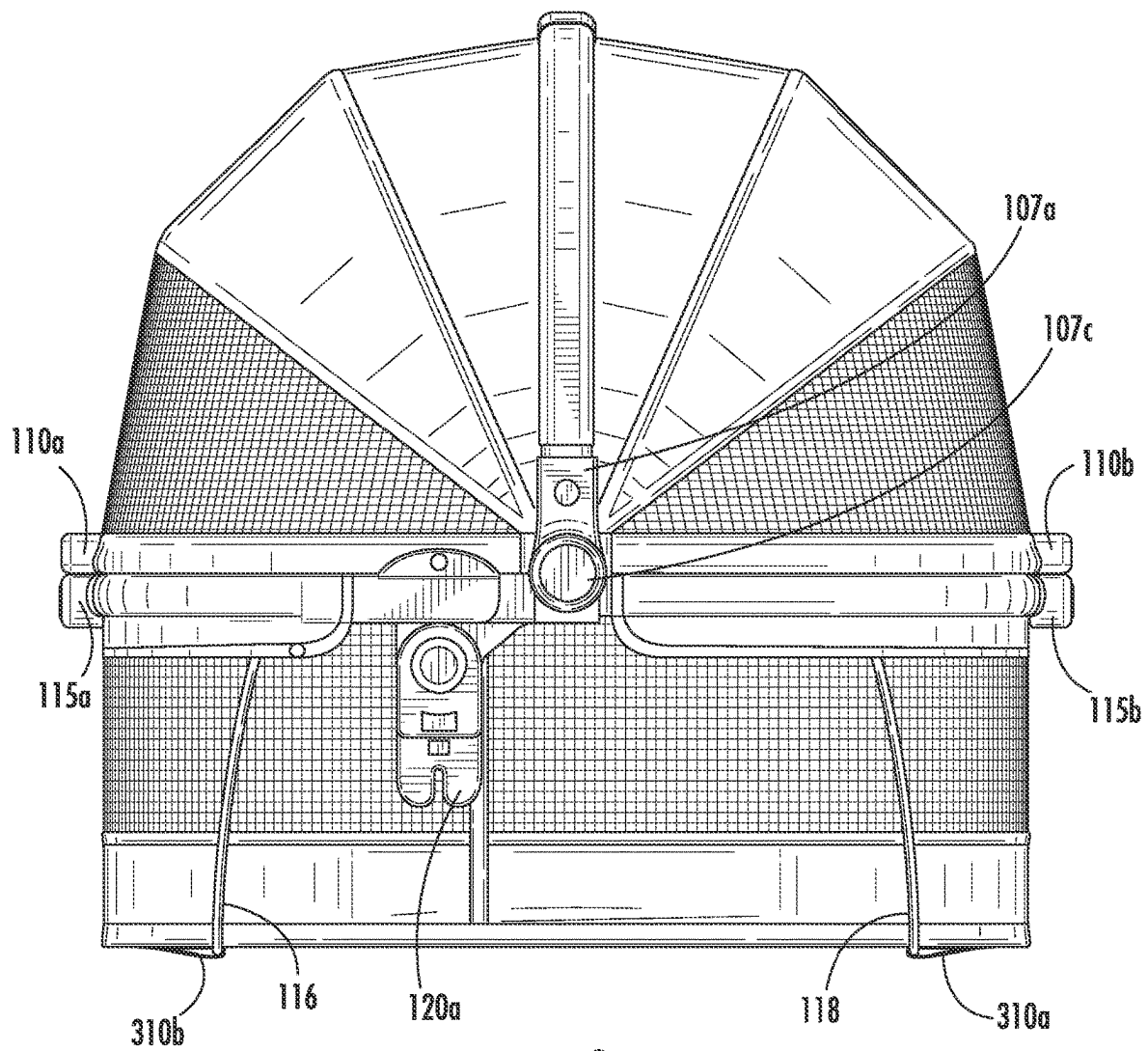


FIG. 3

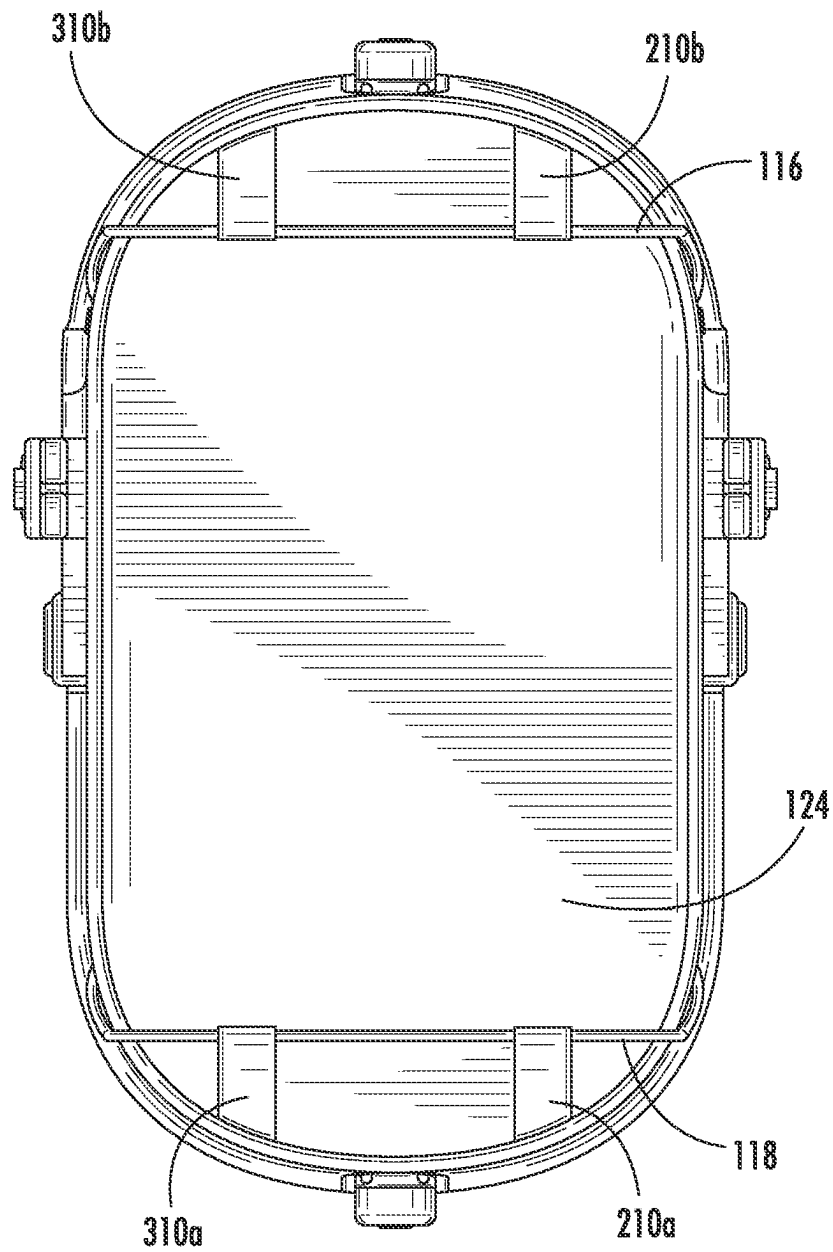


FIG. 4

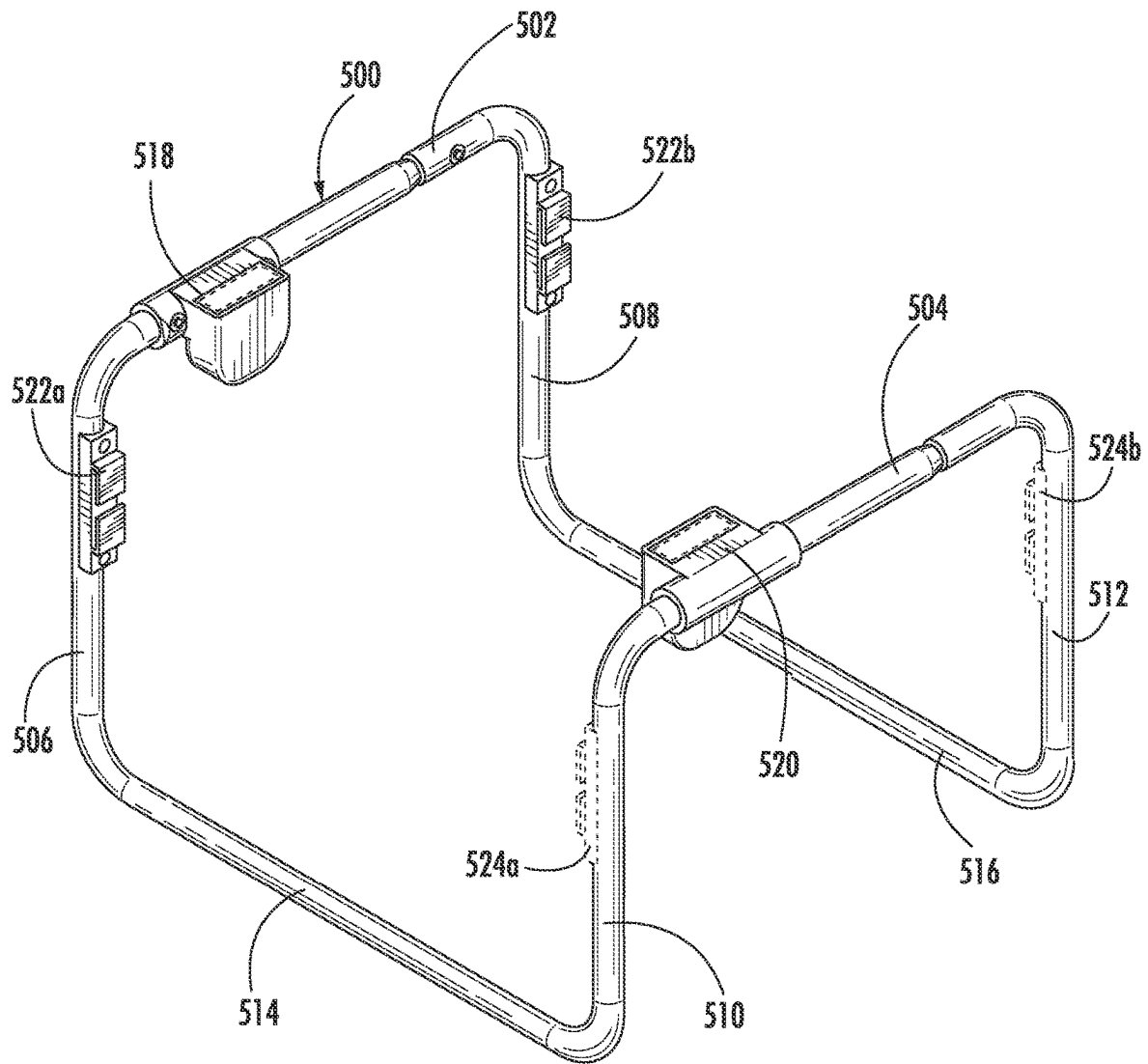


FIG. 5

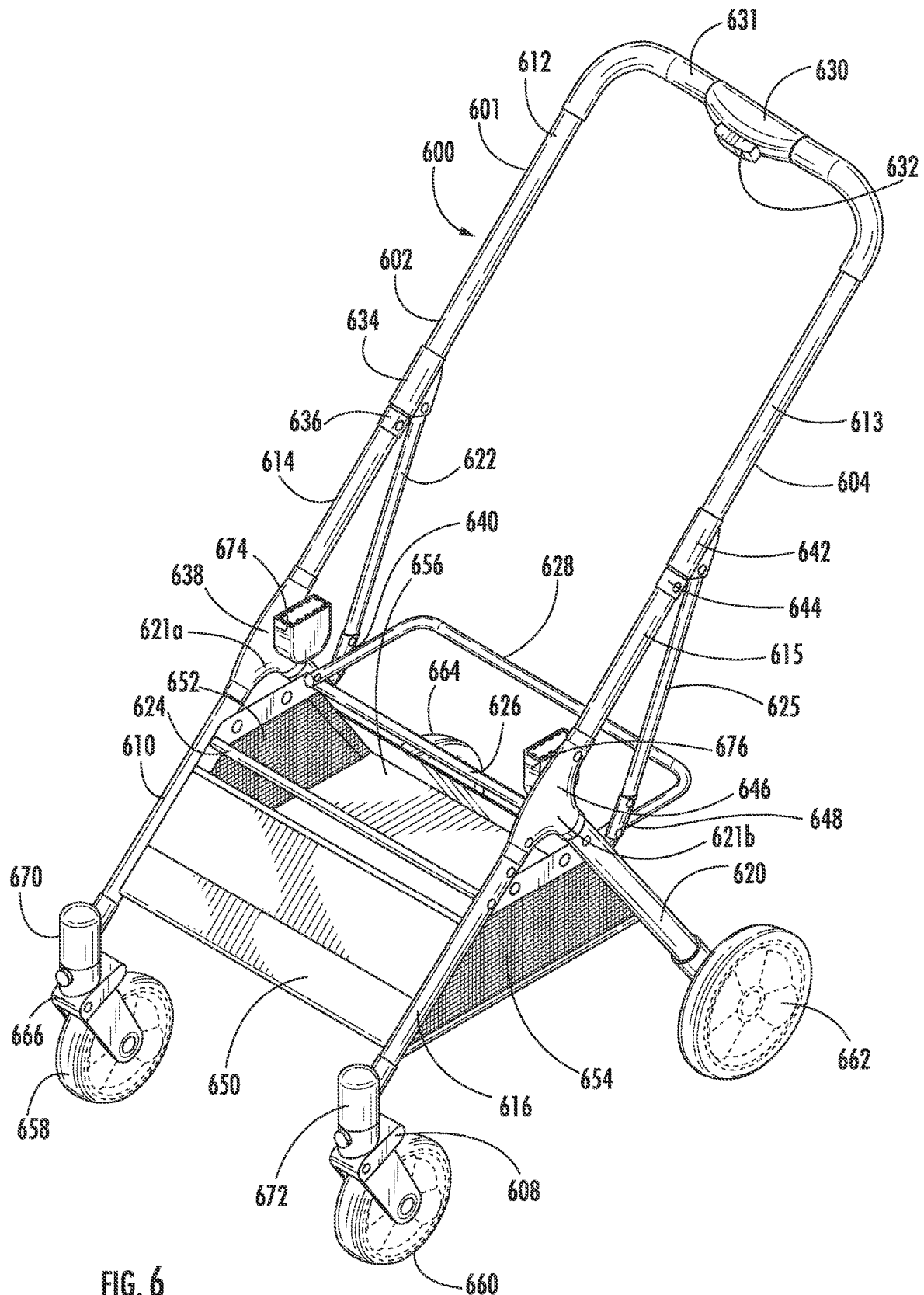


FIG. 6

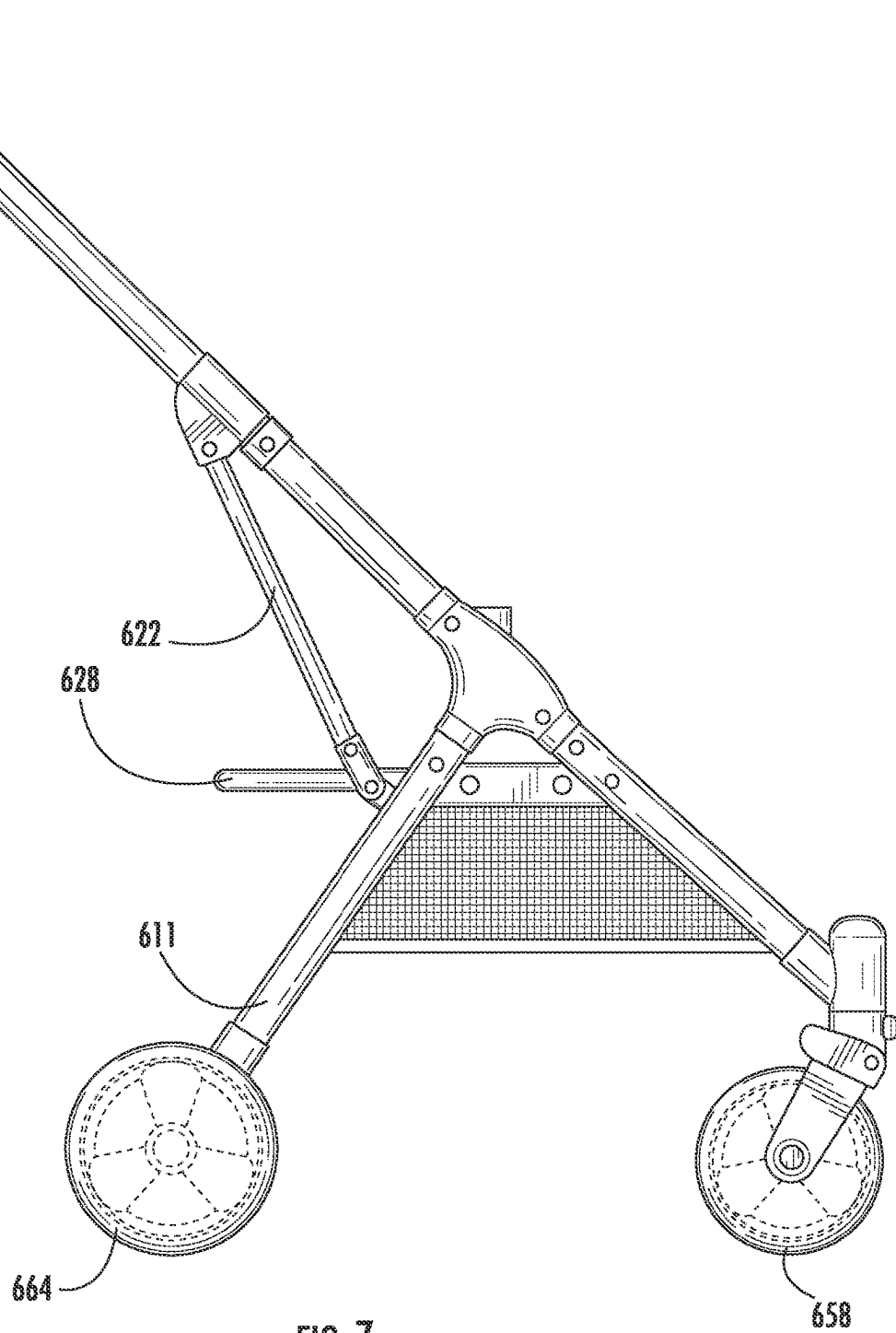


FIG. 7

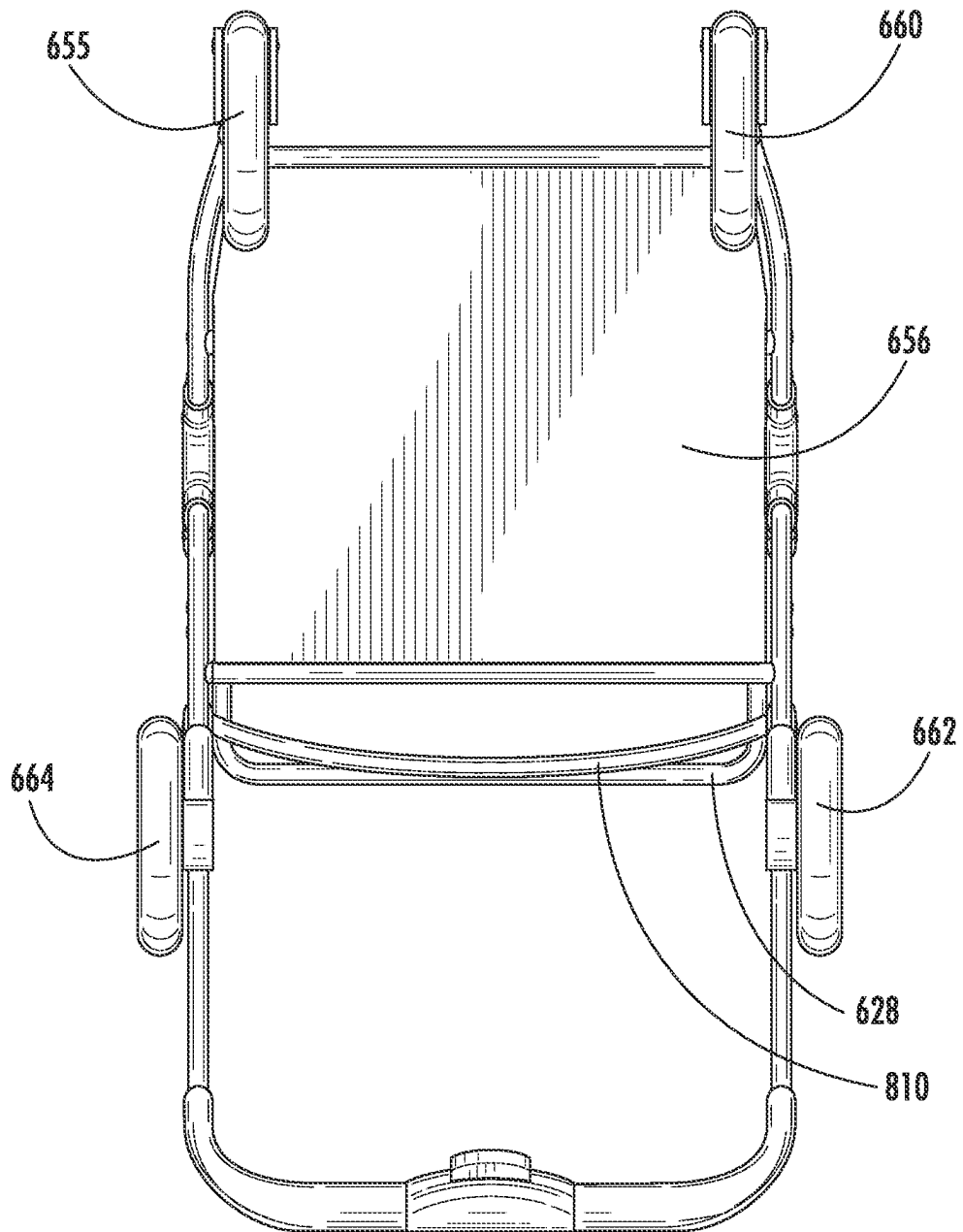


FIG. 8

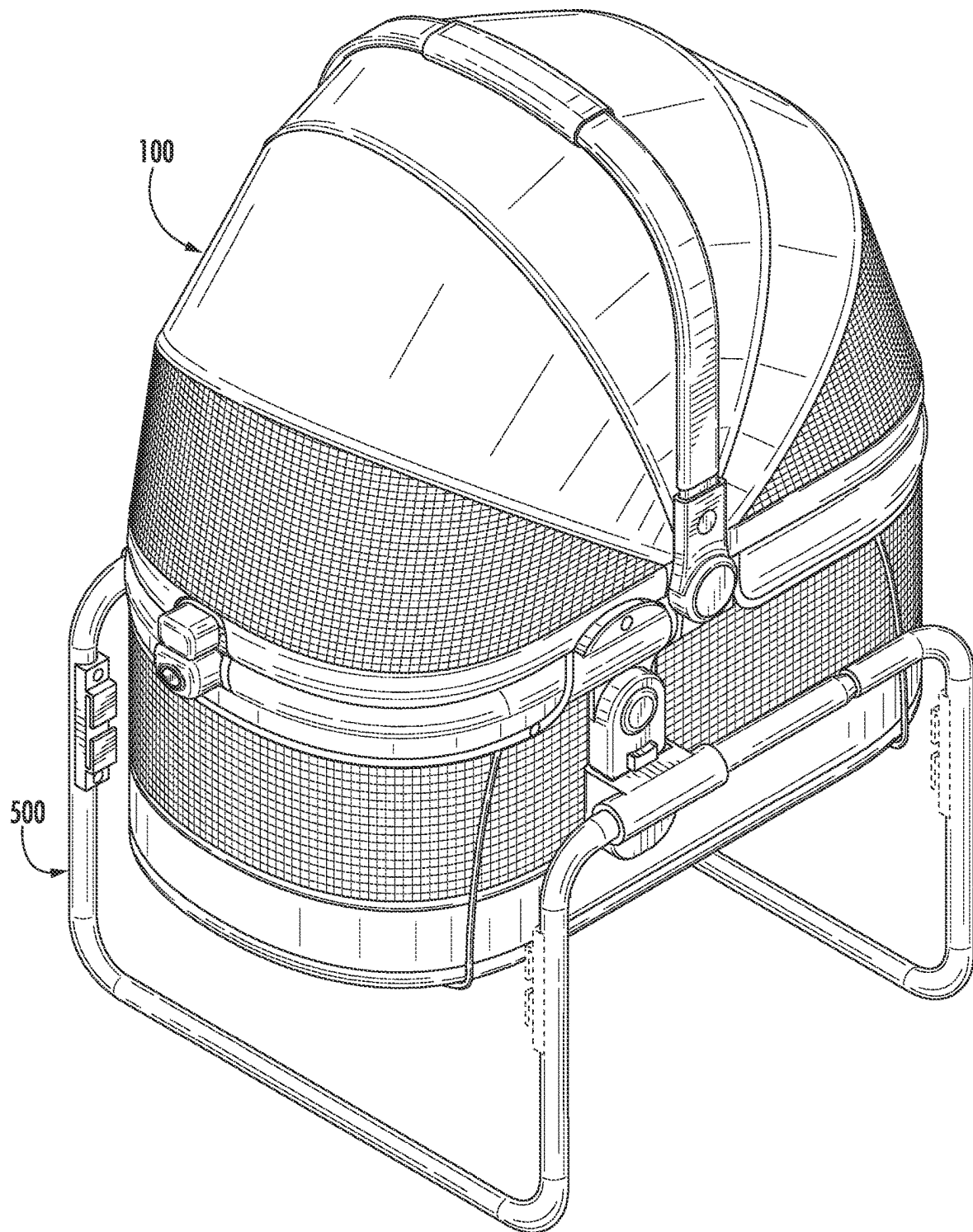
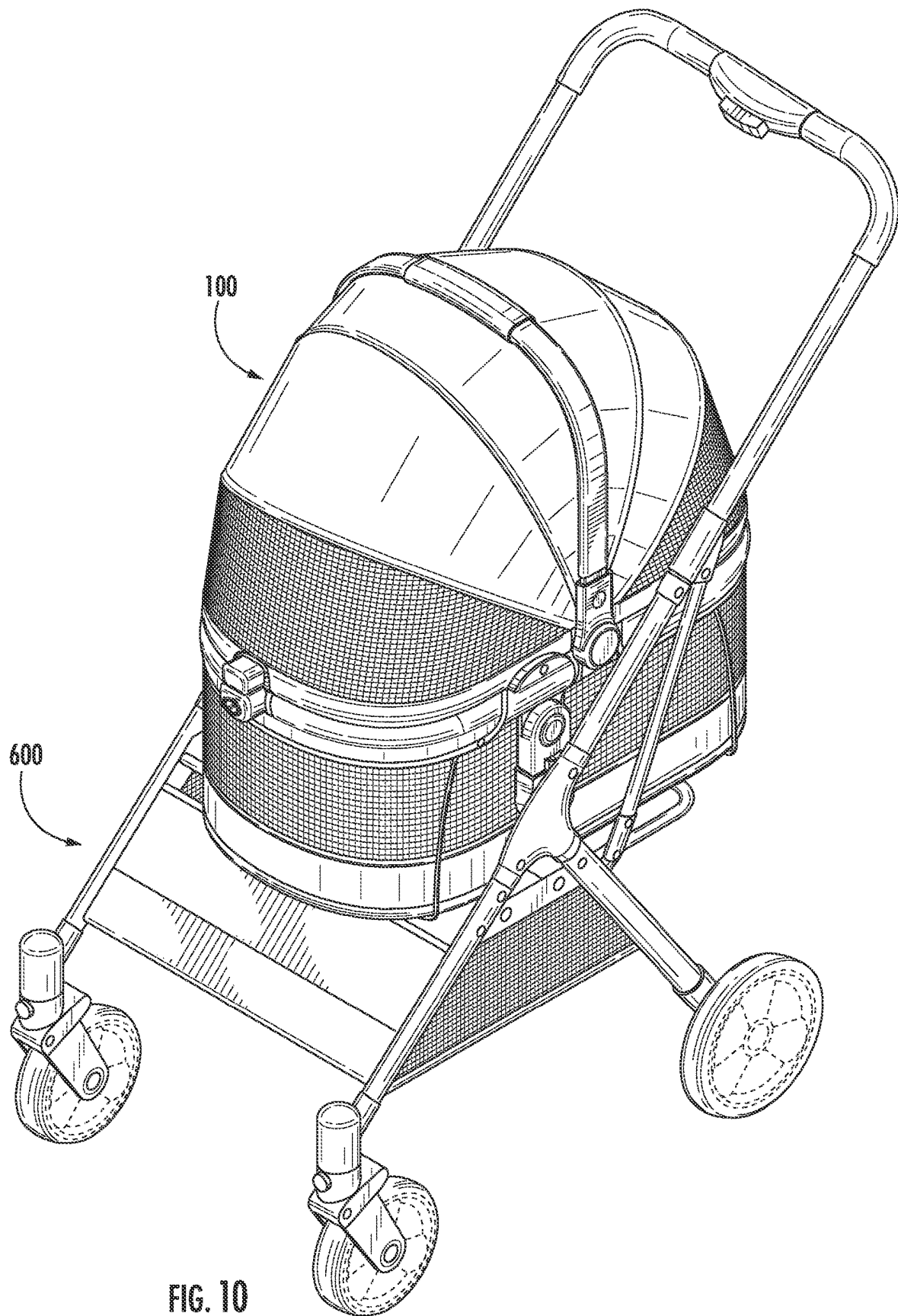


FIG. 9



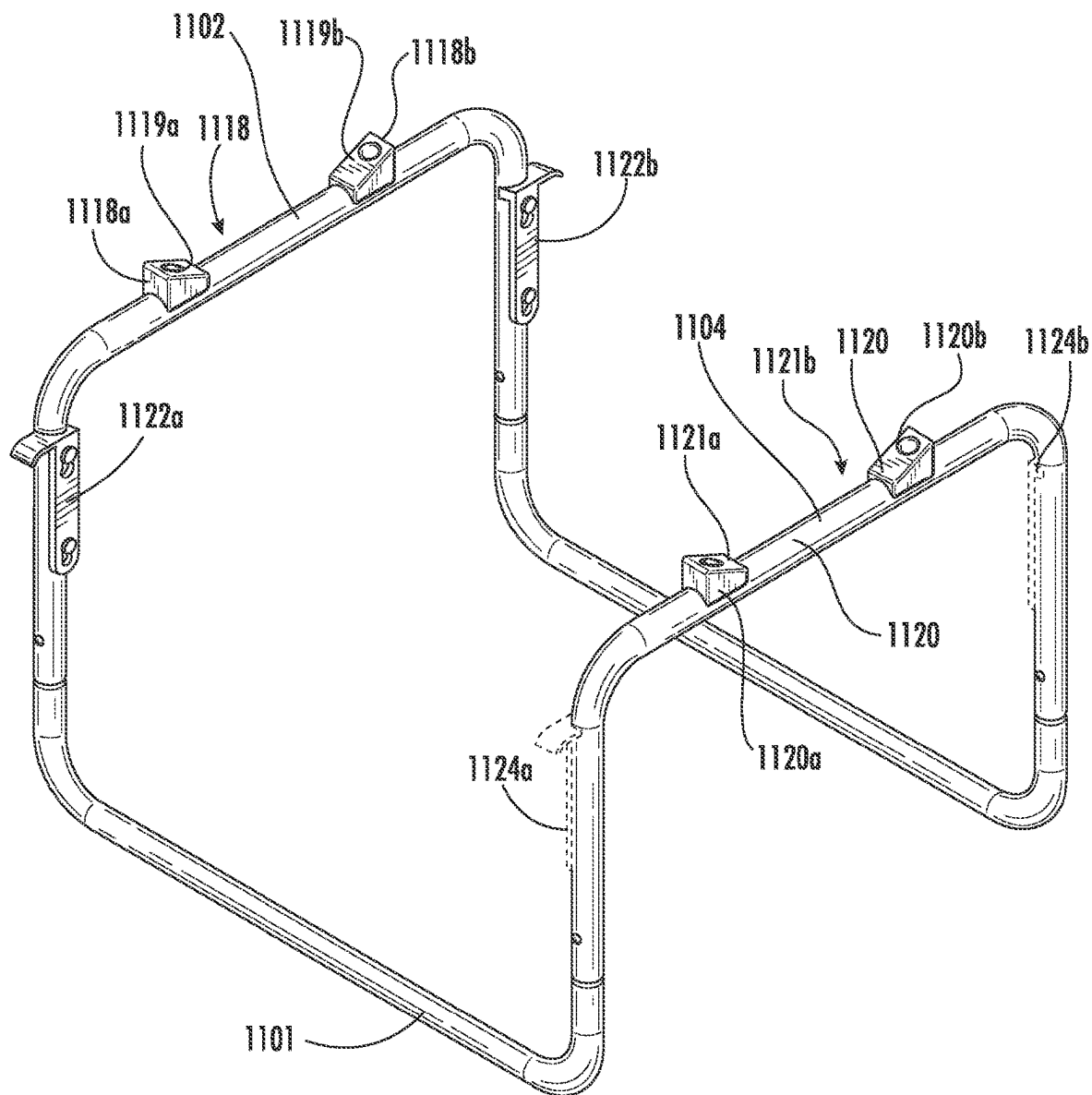


FIG. 11

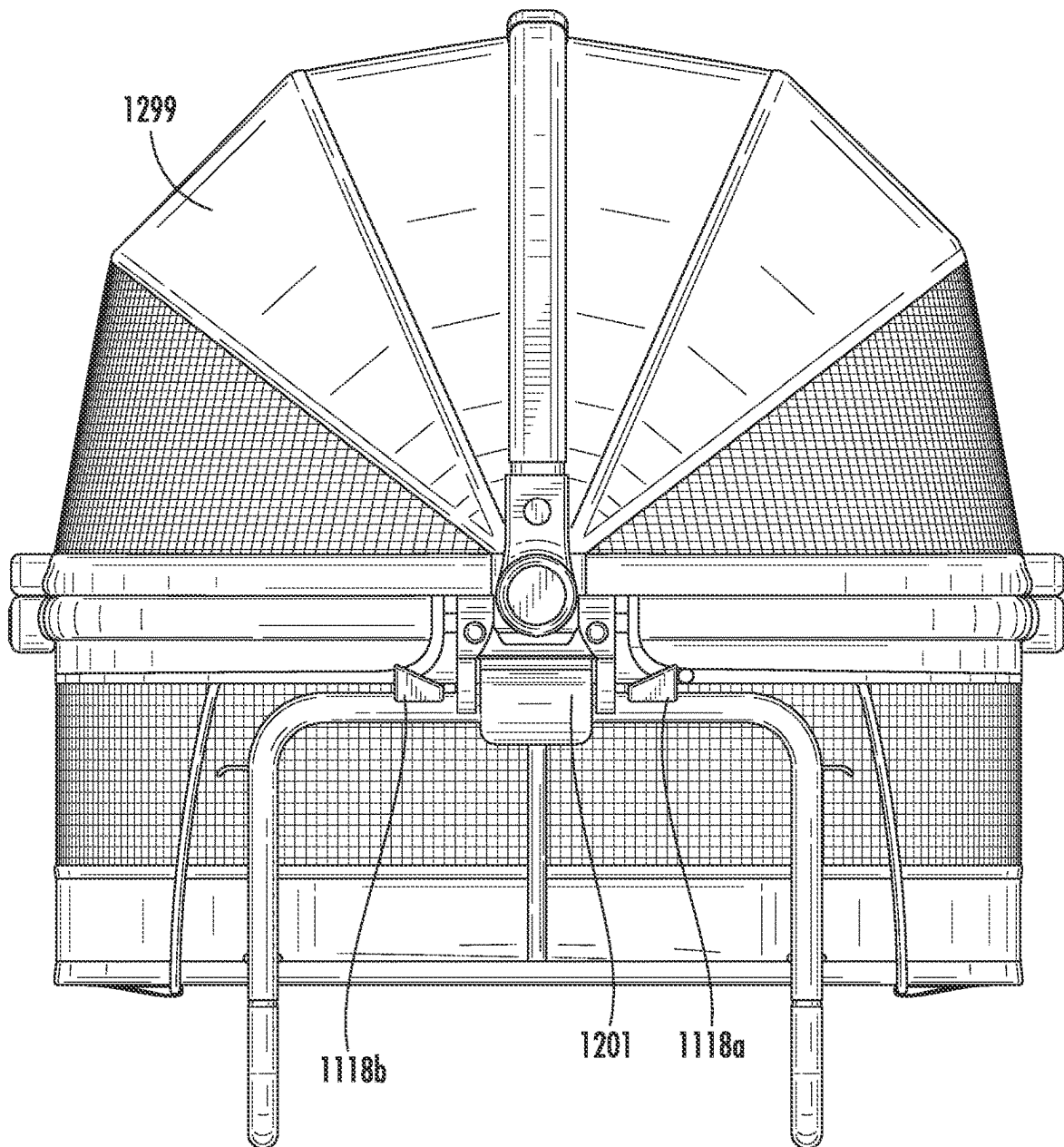


FIG. 12

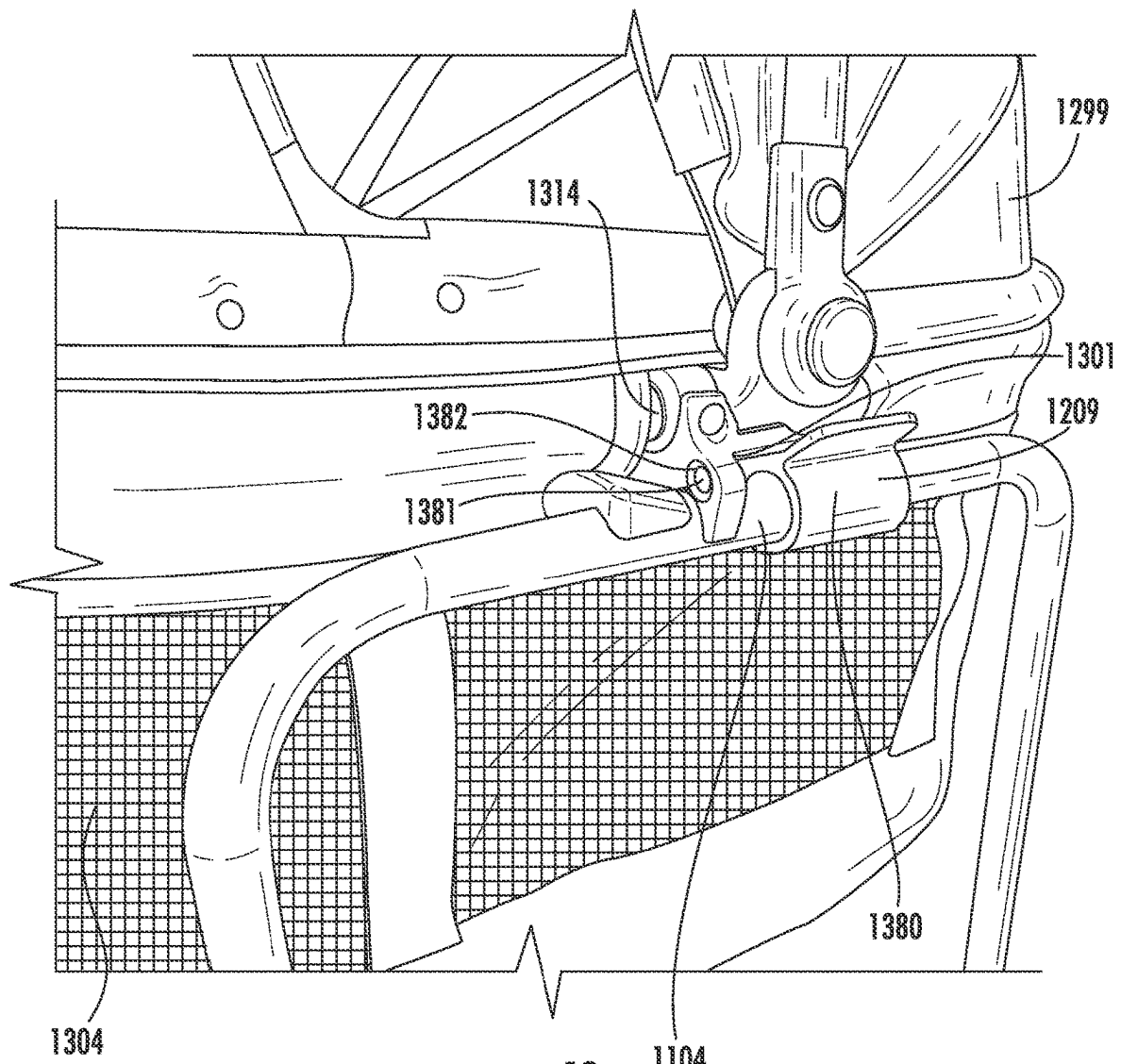


FIG. 13

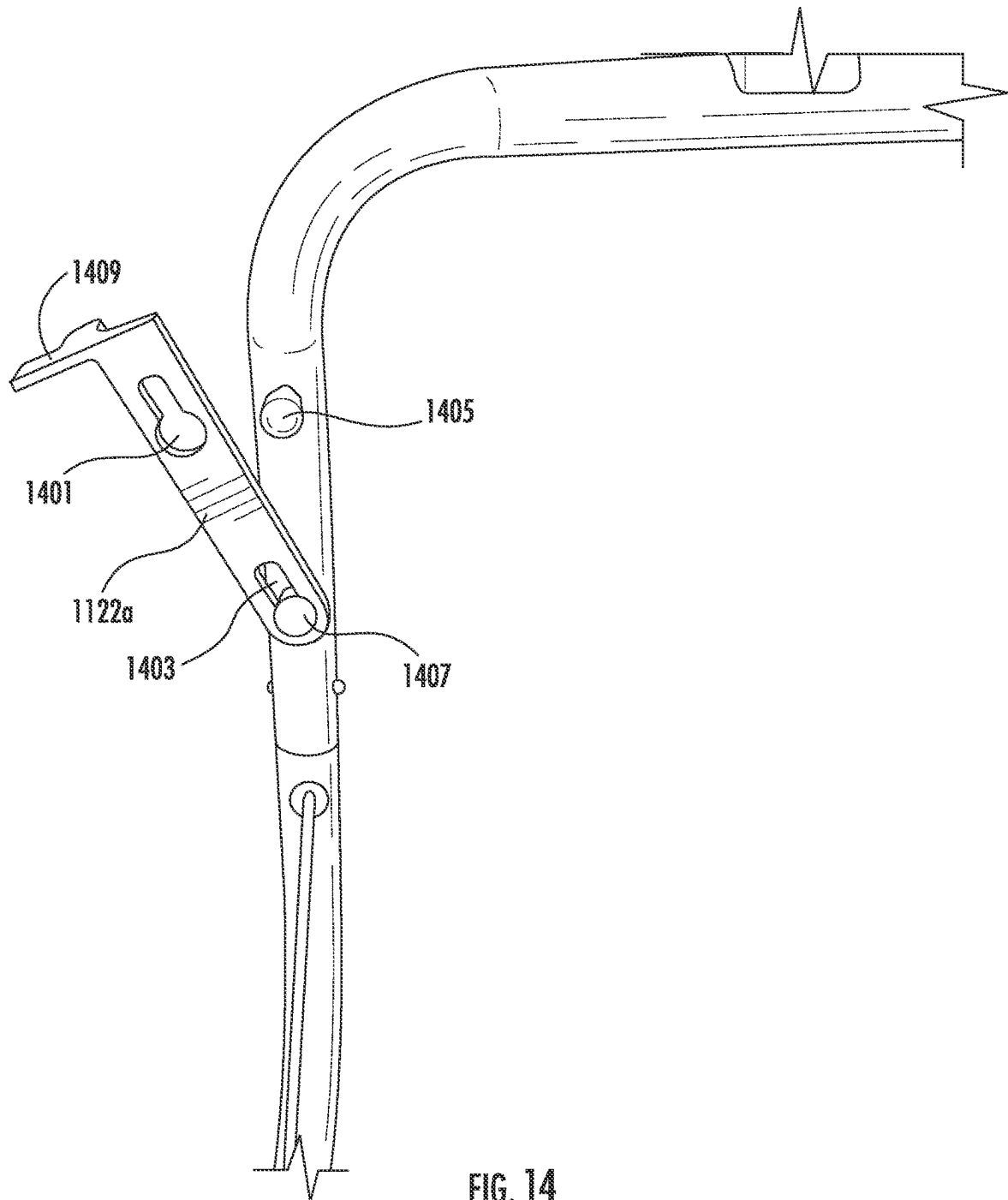


FIG. 14

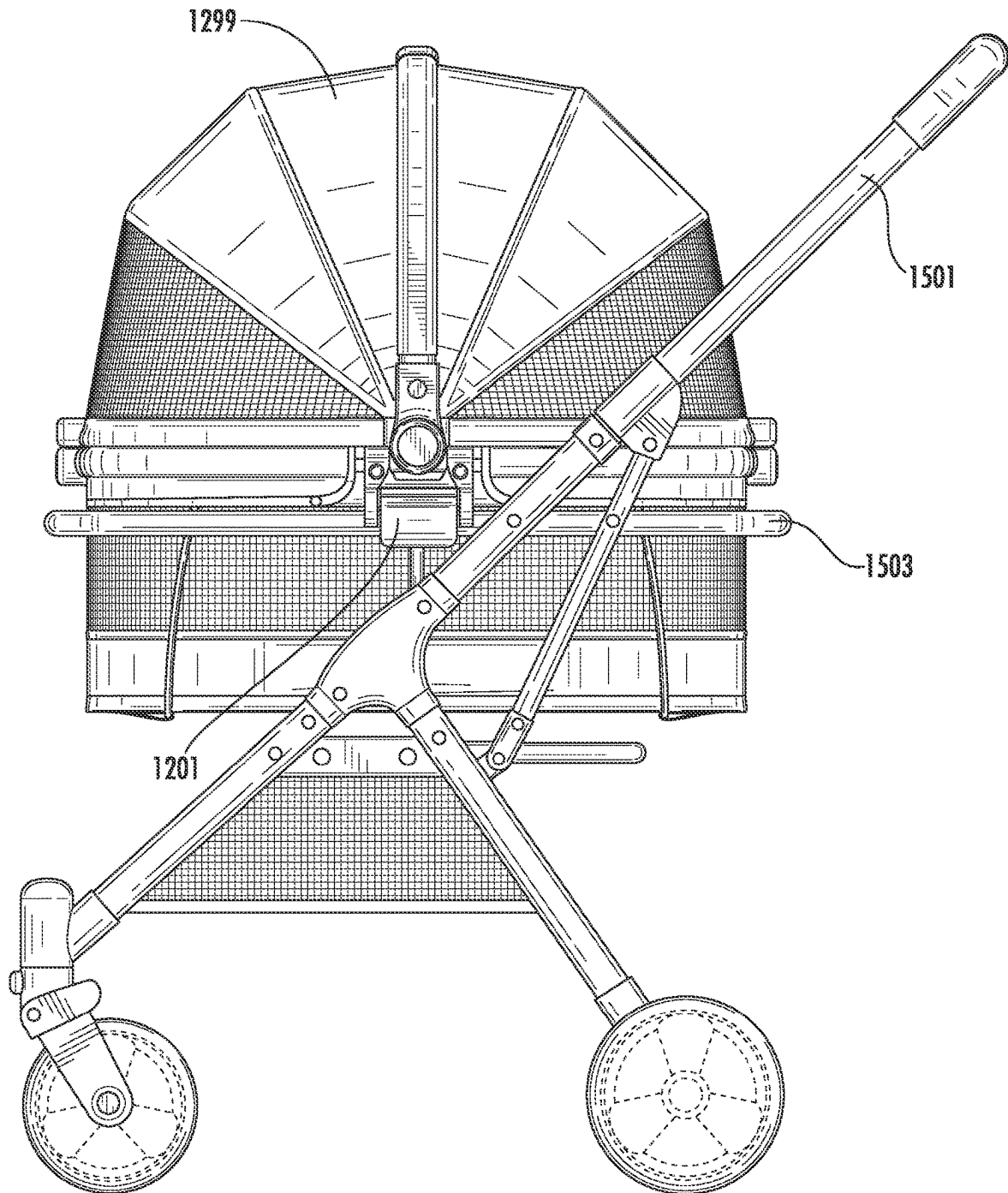


FIG. 15

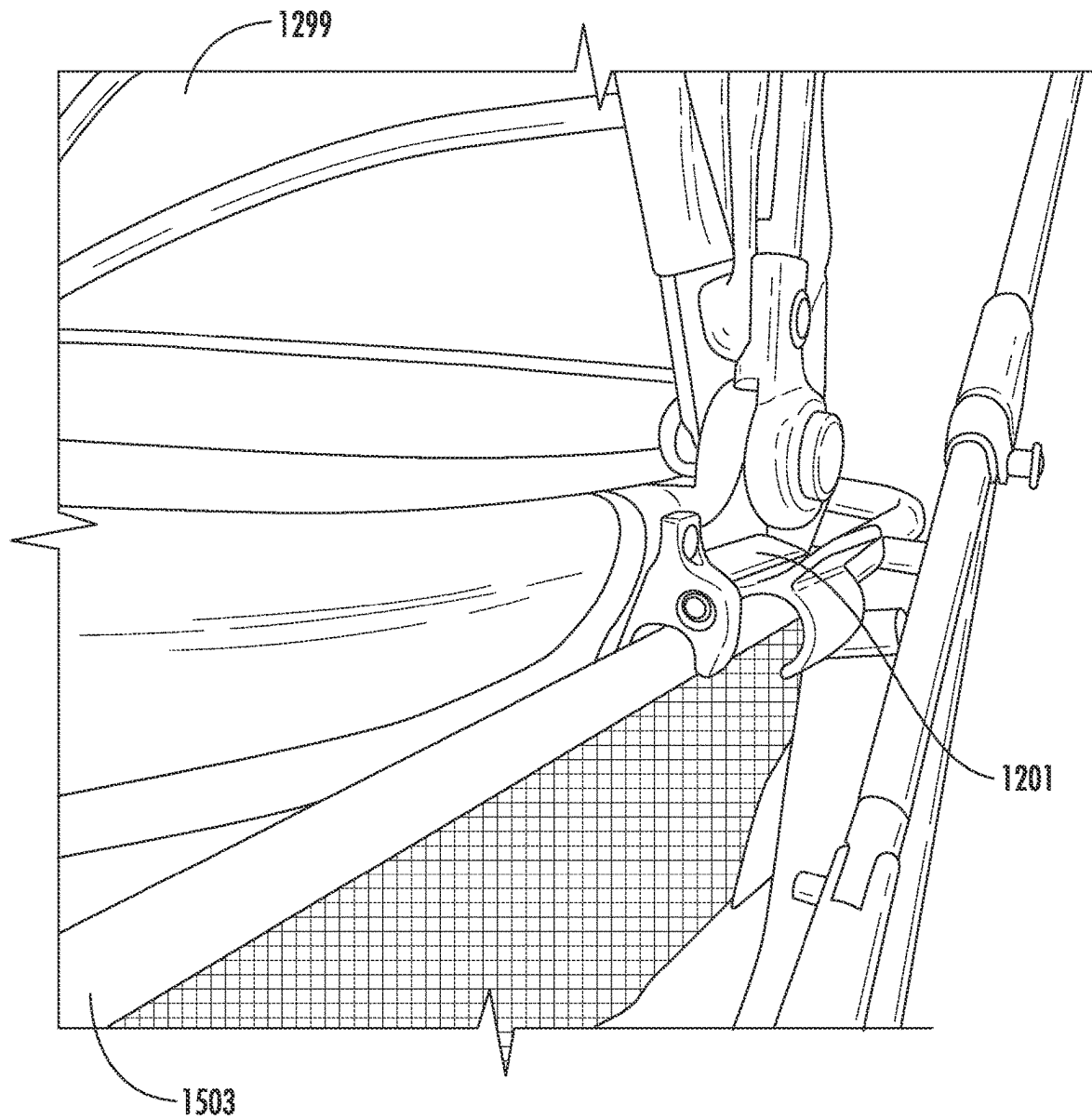


FIG. 16

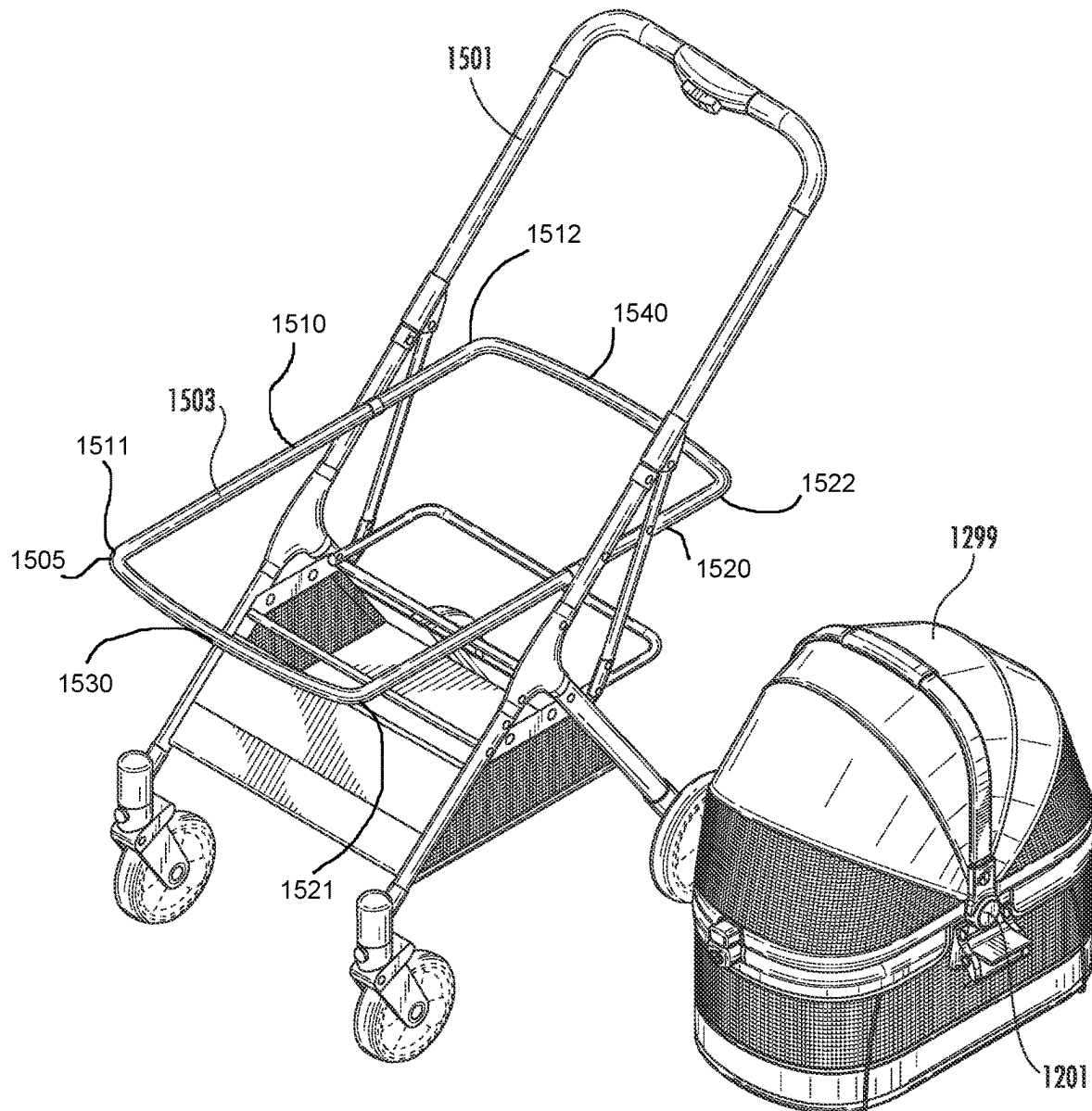


FIG. 17

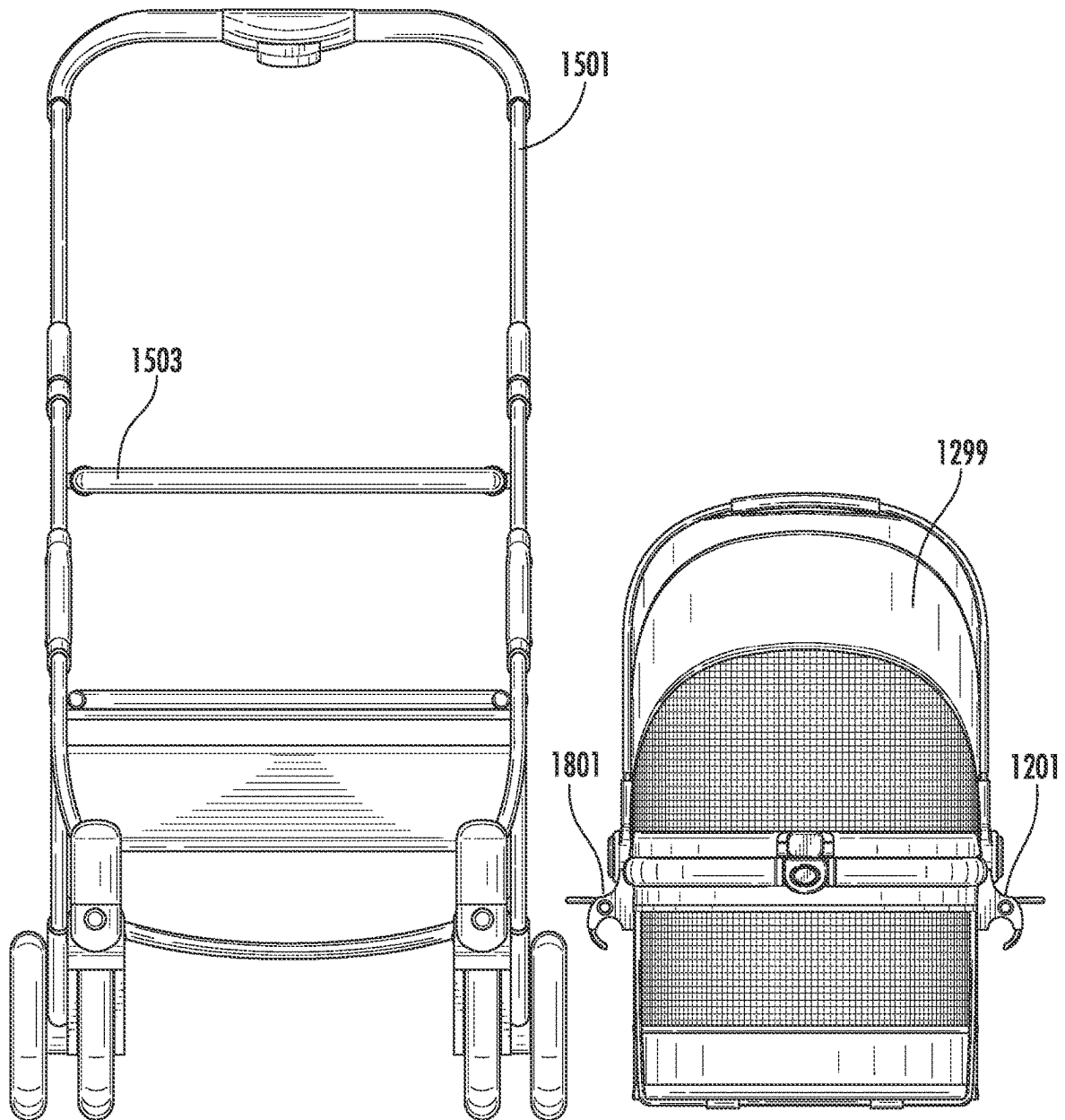
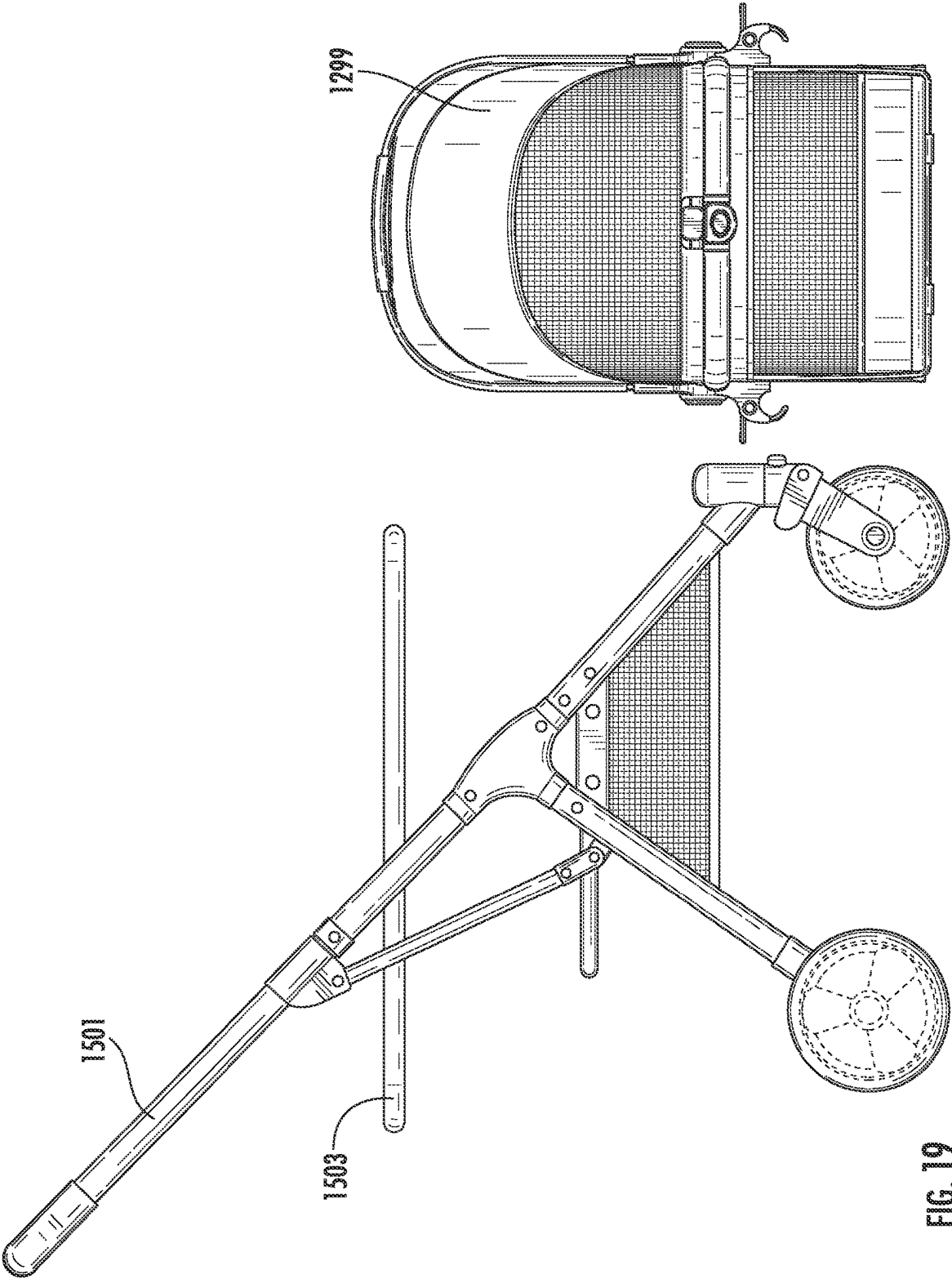


FIG. 18



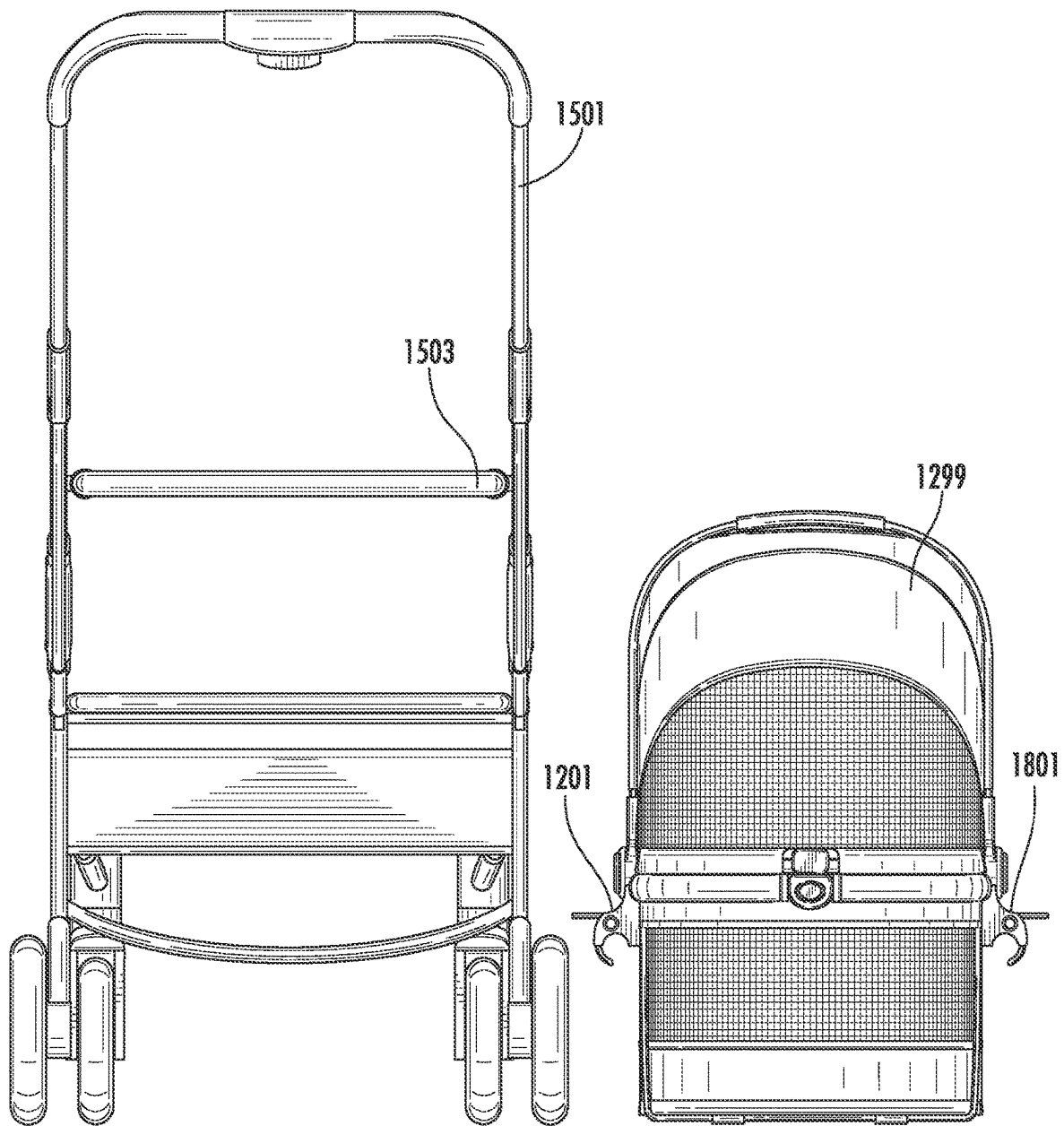


FIG. 20

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CONVERTIBLE PET STROLLER AND CARRIER**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. patent application Ser. No. 16/823,544, filed Mar. 19, 2020, which claims the benefit of U.S. Patent Application No. 62/820,594, filed Mar. 19, 2019, which are incorporated by reference as if fully set forth.

FIELD

The disclosure relates to pet carriers, car seat frames, pet strollers, and assemblies of pet carriers with car seat frames or pet strollers.

BACKGROUND

Many people enjoy taking their pets with them to a variety of locations. Often the pet may be carried. In such a case, a pet carrier may be useful. Devices to carry pets and transport pets within carriers are desired.

SUMMARY

In an aspect, the invention relates to a pet carrier comprising a top section, a bottom section having an upper rim, a handle, a first pet carrier engagement, a second pet carrier engagement, and an interior defined between the top section and the bottom section. The top section comprises a front, a back, a canopy, at least one canopy frame element supporting the canopy, a front canopy latch element at the front, and a back canopy latch element at the back. The canopy comprising a lower edge configured to be proximal to the upper rim of the bottom section. A first canopy frame element of the at least one canopy frame element extends at least partially around the lower edge of the canopy and has a front section at the front and a back section at the back. The front section and the back section meet at a first pivot on a first side of the canopy and meet at a second pivot on a second side of the canopy. The front canopy latch element is fixed to the front section and the back canopy latch element is fixed to the back section. The bottom section comprises a base, a front, a back, a first side, a second side, a shell, a bottom frame element, a front bottom latch element on the front, and a back bottom latch element on the back. The shell extends up from the base to the bottom frame element. The bottom frame element extends around the upper rim of the bottom, and the front and the back bottom latch elements are fixed to the bottom frame element. The handle comprises a first handle receiver engaged with one or more of the first canopy frame element and the first bottom frame element on a first side of the pet carrier. The handle also comprises a second handle receiver engaged with one or more of the first canopy frame element and the first bottom frame element on a second side of the pet carrier. The handle also comprises a handle bar extending from the first handle receiver to the second handle receiver and over the canopy. The front canopy latch element is engageable with the front bottom latch element to secure the front of the top to the front of the bottom in a front closed position, and is releasable to an open position, allowing movement of the top section at the front of the pet carrier to reveal the interior of the pet carrier. The back canopy latch element is engageable with the back bottom latch element to secure the back of the top to the back

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of the bottom in a back closed position, and is releasable to an open position, allowing movement of the top section at the back of the pet carrier to reveal the interior of the pet carrier. The first pet carrier engagement is on the first side of the bottom section. The second pet carrier engagement is on the second side of the bottom section. Both the first pet carrier engagement and the second pet carrier engagement are engageable with a secondary device in order to fix the pet carrier to the secondary device. The top section, the bottom section, and the base form an enclosure, in which a pet may be contained, within the interior of the pet carrier when the front and the back of the canopy are in the closed position.

In an aspect, the invention relates to a car seat frame. The car seat frame comprises a first side, a second side, a first base element having a first end and a second end, and a second base element having a first end and a second end. The first side comprises a first side carrier support, a first vertical support on the first side, and a second vertical support on the first side. Each vertical support on the first side having a first end and a second end, the first end of each fixed to opposite ends of the first side carrier support. The second side comprising a second side carrier support, a first vertical support on the second side, and a second vertical support on the second side. Each vertical support on the second side having a first end and a second end, the first end of each fixed to opposite ends of the second side carrier support. The second end of the first vertical support on the first side is fixed to the first end of the first base element. The second end of the second vertical support on the first side is fixed to the first end of the second base element. The second end of the first vertical support on the second side is fixed to the second end of the first base element. The second end of the second vertical support on the second side is fixed to the second end of the second base element. The car seat frame further comprises a first frame-carrier receiver fixed to the first side carrier support and a second car seat frame-carrier receiver fixed to the second side carrier support.

In an aspect, the invention relates to an assembly comprising a pet carrier and the car seat frame as summarized above.

In an aspect the invention relates to a pet stroller comprising a frame and wheels. The frame comprises a first side, a second side, a push bar, a carrier support. Each side comprises a front lower portion, a rear lower portion, a mid portion, and an upper portion. Each of the front and rear lower portions having first end and a wheel engagement at the first end to which one of the wheels is engaged. The front and rear lower portions on the first side each have a second end opposite the respective first ends that converge at a first junction. The front and rear lower portions on the second side each have a second end opposite the respective first end that converges at a second junction. The mid portion on the first side has a first end at the first junction and a second end extended in a direction generally opposite to the direction in which the lower portion on the first side extends from the first junction. The mid portion on the second side has a first end at the second junction and a second end extending in a direction generally opposite to the direction in which the lower portion on the second side extends from the second junction. The upper portion on the first side has a first end extending from the second end of the mid portion on the first side and a second end. The upper portion on the second side has a first end extending from the second end of the mid portion on the second side and a second end. The push bar having a first end connected to the second end of the upper portion on the first side, and a second end connected to the

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second end of the upper portion on the second side. The carrier support is configured to receive a pet carrier.

In an aspect, the invention relates to an assembly comprising a pet carrier and the pet stroller summarized above.

In an aspect, the invention relates to a system comprising at least two of the above summarized pet carrier, car seat frame, and pet stroller.

In an aspect, the invention relates to a pet carrier. The pet carrier comprises a top section, a bottom section having an upper rim, a handle, a first pet carrier engagement, a second pet carrier engagement, and an interior defined between the top section and the bottom section. The top section comprises a front, a back, a canopy, at least one canopy frame element supporting the canopy, a front canopy latch element at the front, and a back canopy latch element at the back. The canopy comprises a lower edge configured to be proximal to the upper rim of the bottom section, a first canopy frame element of the at least one canopy frame element extending at least partially around the lower edge of the canopy and having a front section at the front and a back section at the back. The front section and the back section meet at a first pivot on a first side of the canopy. The front section and the back section also meet at a second pivot on a second side of the canopy. The front canopy latch element is fixed to the front section, and the back canopy latch element is fixed to the back section. The bottom section comprises a base, a front, a back, a first side, a second side, a shell, a bottom frame element, a front bottom latch element on the front, and a back bottom latch element on the back. The shell extends up from the base to the bottom frame element. The bottom frame element extends around the upper rim of the bottom. The front and the back bottom latch elements are fixed to the bottom frame element. The handle comprising a first handle receiver engaged with one or more of the first canopy frame element and the first bottom frame element on a first side of the pet carrier, and a second handle receiver engaged with one or more of the first canopy frame element and the first bottom frame element on a second side of the pet carrier. The handle also comprises a handle bar extending from the first handle receiver to the second handle receiver and over the canopy. The front canopy latch element being engageable with the front bottom latch element to secure the front of the top to the front of the bottom in a front closed position. The front canopy latch element also being releasable to an open position, allowing movement of the top section at the front of the pet carrier to reveal the interior of the pet carrier. The back canopy latch element being engageable with the back bottom latch element to secure the back of the top to the back of the bottom in a back closed position. The back canopy latch element also being releasable to an open position, allowing movement of the top section at the back of the pet carrier to reveal the interior of the pet carrier. The first pet carrier engagement on the first side of the bottom section. The second pet carrier engagement on the second side of the bottom section. Both the first pet carrier engagement and the second pet carrier engagement being engageable with respective receivers on a secondary device in order to fix the pet carrier to the secondary device. The top section, the bottom section, and the base forming an enclosure, in which a pet may be contained, within the interior of the pet carrier when the front and the back of the canopy are in the closed position.

In another aspect, the invention relates to a car seat frame. The car seat frame comprises a first side, a second side, a first base element having a first end and a second end, and a second base element having a first end and a second end. The first side comprises a first side carrier support, a first

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vertical support on the first side, and a second vertical support on the first side. Each vertical support on the first side has a first end and a second end, the first end of each fixed to opposite ends of the first side carrier support. The second side comprises a second side carrier support, a first vertical support on the second side, and a second vertical support on the second side. Each vertical support on the second side having a first end and a second end, the first end of each fixed to opposite ends of the second side carrier support. The second end of the first vertical support on the first side is fixed to the first end of the first base element. The second end of the second vertical support on the first side is fixed to the first end of the second base element. The second end of the first vertical support on the second side is fixed to the second end of the first base element, and the second end of the second vertical support on the second side is fixed to the second end of the second base element. The car seat frame further comprises a first frame-carrier receiver fixed to the first side carrier support and a second car seat frame-carrier receiver fixed to the second side carrier support. The car seat frame may comprise one or more seat belt retainers, where each seat belt retainer is fixed to one of the vertical supports. The one or more seat belt retainers may comprise a first seat belt retainer fixed to the first vertical support on the first side, and a second seat belt retainer fixed to the second vertical support on the first side. The car seat frame may comprise a first frame-carrier receiver fixed to the first side carrier support and a second frame-carrier receiver fixed to the second side carrier support.

In another aspect, the invention relates to an assembly comprising a pet carrier and the car seat frame. The pet carrier may be the pet carrier summarized above. When the pet carrier is the pet carrier summarized above, the first pet carrier engagement may be engaged with the first frame-carrier receiver, and the second pet carrier engagement may be engaged with the second frame-carrier receiver.

In an aspect, the invention relates to a pet stroller comprising a frame and four wheels. The frame comprises a first side, a second side, a push bar, a pet carrier support. Each side comprises a front lower portion, a rear lower portion, a mid portion, and an upper portion. Each of the front and rear lower portions having a first end and a wheel engagement at the first end to which one of the wheels is engaged. The front and rear lower portions on the first side each have a second end opposite the respective first ends that converge at a first junction. The front and rear lower portions on the second side each have a second end opposite the respective first end that converge at a second junction. The mid portion on the first side has a first end at the first junction and a second end extended in a direction generally opposite to the direction in which the lower portion on the first side extends from the first junction. The mid portion on the second side has a first end at the second junction and a second end extending in a direction generally opposite to the direction in which the lower portion on the second side extends from the second junction. The upper portion on the first side has a first end extending from the second end of the mid portion on the first side and a second end. The upper portion on the second side has a first end extending from the second end of the mid portion on the second side and a second end. The push bar has a first end connected to the second end of the upper portion on the first side, and the second end connected to the second end of the upper portion on the second side.

In another aspect, the invention relates to an assembly comprising a pet carrier and the pet stroller summarized above. The pet carrier may rest upon the front and rear carrier supports. The pet carrier may be the pet carrier

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summarized above. When the pet carrier is the pet carrier summarized above, the first and second pet carrier engagements may be respectively engaged with the first and second pet carrier receivers.

In an aspect, the invention relates to a system comprising at least two of the above summarized aspects.

BRIEF DESCRIPTION OF THE DRAWINGS

The following detailed description of embodiments of the present invention will be better understood when read in conjunction with the appended drawings. For the purpose of illustrating the invention, there are shown in the drawings embodiments which are presently preferred. It is understood, however, that the invention is not limited to the precise arrangements and instrumentalities shown. In the drawings:

FIG. 1 illustrates a perspective view of a pet carrier.

FIG. 2 illustrates a first side view of the pet carrier.

FIG. 3 illustrates a second side view of the pet carrier.

FIG. 4 illustrates a bottom view of the pet carrier.

FIG. 5 illustrates a car seat frame.

FIG. 6 illustrates a pet stroller.

FIG. 7 illustrates a side view of the pet stroller.

FIG. 8 illustrates a bottom view of the pet stroller.

FIG. 9 illustrates an assembly of a pet carrier and a car seat frame.

FIG. 10 illustrates an assembly of a pet carrier and a pet stroller.

FIG. 11 illustrates a car seat frame exemplifying alternate frame-carrier receivers and alternate seat belt retainers.

FIG. 12 illustrates a pet carrier with pet carrier engagements in combination with the car seat frame of FIG. 11.

FIG. 13 illustrates one of the pet carrier engagements on the pet carrier illustrated in FIG. 12 engaged with one of the frame-carrier receivers of the car seat frame of FIG. 11.

FIG. 14 illustrates one of the seat belt retainers on the car seat frame of FIG. 11.

FIG. 15 illustrates a pet carrier having pet carrier engagements in combination with a stroller.

FIG. 16 illustrates the combination of FIG. 15 with one of the pet carrier engagements in an open position but on the stroller.

FIG. 17 illustrates a perspective view of the pet carrier and the stroller of FIG. 15 in an un-combined state.

FIG. 18 illustrates a front view of the pet carrier and the stroller of FIG. 15 in an un-combined state.

FIG. 19 illustrates a side view of the pet carrier and the stroller of FIG. 15 in an un-combined state.

FIG. 20 illustrates a rear view of the pet carrier and the stroller of FIG. 15 in an un-combined state.

DETAILED DESCRIPTION

Certain terminology is used in the following description for convenience only and is not limiting. The words "right," "left," "top," and "bottom" designate directions in the drawings to which reference is made.

The words "a" and "one," as used in the claims and in the corresponding portions of the specification, are defined as including one or more of the referenced item unless specifically stated otherwise. The phrase "at least one" followed by a list of two or more items, such as "A, B, or C," means any individual one of A, B or C as well as any combination thereof.

An embodiment includes a pet carrier. Referring to FIGS. 1-4, an embodiment is illustrated as pet carrier 100, which may comprise a top section 102, a bottom section 104

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having an upper rim 125, a handle 106, a first pet carrier engagement 120a, a second pet carrier engagement 120b, and an interior defined between the top section and the bottom section.

The top section 102 may comprise a front, a back, and a canopy 108. The top section 102 may also comprise at least one canopy frame element supporting the canopy, a front canopy latch element 110a at the front, and a back canopy latch element 110b at the back, as illustrated for top section 102. Ridges 111a, 111b, 111c, and 111d may be present, and may be seams of different panels of the canopy, and/or engagement areas between the canopy at frame elements inside the canopy. Frame elements inside the canopy 108 may be enclosed within or tethered to canopy material. The canopy 108 may also comprise a lower edge 109 configured to be proximal to the upper rim 125 of the bottom section 104. A first canopy frame element 111 of the at least one canopy frame element having a front section 126 and a back section 127 may extend at least partially around the lower edge of the canopy. The front section 126 and the back section 127 may meet at a first pivot 128 on a first side of the canopy. The front section 126 and the back section 127 may also meet at a second pivot 129 on a second side of the canopy. The front section 126, back section 127, first pivot 128, and second pivot configured such that one or both of the front and back sides of the canopy may be rotated upward or downward. The front canopy latch element 110a may be attached to the front section 126 of the first canopy frame element 111, and the back canopy latch element 110b may be fixed to the back section 127 of the first canopy frame element 111. The frame elements other than the first canopy frame element 111 may each comprise a first end associated with an attachment point on the first side of the pet carrier, and a second end associated with an attachment point on the second side of the pet carrier. The attachment points may be at the first pivot 128 and the second pivot 129 such that the one or both of the front and back sides of the canopy may be rotated upward or downward. Intermediate portions, between the first and second ends, of these frame elements may be configured to contact, be contained within the material of, or be tethered to the canopy 108.

The pet carrier 100 has the bottom section 104, which may comprise a base 124, a front, a back, a first side, a second side, a shell 105, at least one bottom frame element, a front bottom latch element 115a on the front, and a back bottom latch element 115b on the back. A first bottom frame element 114 of the at least one bottom frame element may extend around the upper rim 125 of the bottom section 104. The front and the back bottom latch elements (115a, 115b) may be fixed to the first bottom frame element 114.

The material of a portion or the whole of at least one of the canopy or the shell may be opaque, translucent, solid, mesh. Any type of material may be used. The material may be nylon.

A pet carrier may include a handle, which may be the handle 106 illustrated. The handle 106 may comprise a handle bar 130, a first handle receiver 107a, and a second handle receiver 107b. The first handle receiver 107a may be engaged with at least one of the first canopy frame element 111 or the first bottom frame element 114 on the first side of the pet carrier. The second handle receiver 107b may be engaged with at least one of the first canopy frame element 111 or the first bottom frame element 115b on the second side of the pet carrier. The handle bar 130 may extend from the first handle receiver 107a to the second handle receiver 107b and over the canopy 108. Each handle receiver may be separate from the handle bar 130 and affixed to it, or integral

with the handle bar **130**. Each handle receiver may also be affixed to at least one of the first canopy frame element **111** or the first bottom frame element **114** or integral thereto. In an embodiment, the handle **106** is fixed in position. In an embodiment, each handle receiver may comprise a handle release (**107c**, **107d**), which when activated allow rotation of the handle bar **130** over the top of the pet carrier **100**.

The handle **106** may be comprised of more than one part or be a single piece. The pet carrier may have a grip **131**, that is either a separate piece affixed to the handle bar **130** or integral with the handle bar **130**.

The front canopy latch element **110a** may be engageable with the front bottom latch element **115a** to secure the front of the top section **102** to the front of the bottom section **104** in a front closed position. The front canopy latch element **110a** may also be releasable to an open position, allowing movement of the top section **102** at the front of the pet carrier to reveal the interior of the pet carrier.

The back canopy latch element **110b** may be engageable with the back bottom latch element **115b** to secure the back of the top section **102** to the back of the bottom section **104** in a back closed position. The back canopy latch element **110b** may be capable of being released to an open position, allowing movement of the top section **102** at the back of the canopy to reveal the interior of the pet carrier.

The first pet carrier engagement **120a** is illustrated on the first side of the bottom section **104**, and extending toward the base **124**. The second pet carrier engagement **120b** is illustrated on the second side of the bottom section **104**, and extending toward the base **124**. Both the first pet carrier engagement **120a** and the second pet carrier engagement **120b** may be capable of engaging respective receivers on a secondary device in order to fix the pet carrier to the secondary device. The secondary device may be a car seat frame or a pet stroller. The carrier engagements **120a**, **120b** may be affixed to or integral with the first bottom frame element **114**, or other structures on the pet carrier. In an embodiment, the carrier engagements **120a**, **120b** are at alternate locations on the pet carrier to fit to a specific secondary device. In an embodiment, the carrier engagements **120a**, **120b** may have alternate structures/mechanisms and/or be moved forward or backward along the pet carrier to so that the pet carrier may be fit to various secondary devices.

The top section **102** and the bottom section **104** may form an enclosure, in which a pet may be contained, within the interior of the pet carrier when the front and the back of the canopy are in the closed position. A pet may also be resident within the bottom section **104** while at least one or both of the front and back portions of the top section **102** are raised. The pet may then be able to view the surrounding environment, or others may be able to interact with the pet visually or by touch.

The pet carrier illustrated further comprises a first support wire **116** proximal to the front of the bottom section **104**. The first support wire may be engaged with the first bottom frame element **114** on the first and second sides of the bottom section **104**, and is illustrated extending from the first bottom frame element **114** along the first side of the bottom section **104** toward the base **124**, across the base **124**, and along the second side of the bottom section **104** toward the first bottom frame element **114**. The pet carrier illustrated also further comprises a second support wire **118** proximal to the rear of the bottom section **104**, engaged with the first bottom frame element **114** on the first and second sides of the bottom section **104**, and is illustrated extending from the first bottom frame element **114** along the first side of the

bottom section **104** toward the base **124**, across the base **124**, and along the second side of the bottom section **104** toward the first bottom frame element **114**. The embodiment illustrated shows the support wires exterior to the shell **105**. One or both of the first support wire **116** and second support wire **118** may be positioned exterior to the bottom section **104**, interior to the bottom section **104**, or within the material of the bottom section **104**. The support wires **116**, **118** may stabilize the structure of the pet carrier **100**. The support wires **116**, **118** may be utilized to stabilize the bottom when the pet carrier **100** is set on a surface. The embodiment illustrated includes wire holders **210a**, **210b**, **310a**, and **310b** that secure the support wires to the base **124**, and may stabilize the positioning of the support wires. Embodiments of pet carriers herein may include alternate support structures to stabilize the pet carrier.

An embodiment includes a car seat frame. Referring to FIG. 5, an embodiment is illustrated as car seat frame **500**, which may comprise a first side, a second side, a first base element **514** having a first end and a second end, and a second base element **516** having a first end and a second end.

The first side may comprise a first side carrier support **502**, a first vertical support on the first side **506**, and a second vertical support on the first side **508**. Each of vertical supports **506**, **508** have a first end and a second end, and the first end of each is fixed to opposite ends of the first side carrier support **502**.

The second side may comprise a second side carrier support **504**, a first vertical support on the second side **510**, and a second vertical support on the second side **512**. Each of vertical supports **510**, **512** have a first end and a second end, the first end of each fixed to opposite ends of the second side carrier support **504**.

As illustrated, the second end of the first vertical support on the first side **506** is fixed to the first end of the first base element **514**, and the second end of the second vertical support on the first side **508** is fixed to the first end of the second base element **516**. The second end of the first vertical support on the second side **510** is fixed to the second end of the first base element **514**, and the second end of the second vertical support on the second side **512** is fixed to the second end of the second base element **516**.

FIG. 5 illustrates an embodiment of the car seat frame **500** where the first vertical supports **506**, **510** on the first and second sides are integral with the first base element **514**, and the first vertical supports **506**, **510** are integral with a portion of the first side carrier support **502** and second side carrier support **504**, respectively. This portion of the car seat frame may be referred to as a first piece. The first side carrier support **502** and second side carrier support **504** each include a junction where a non-integral second piece is fixed and extends to the second end of each of the first side carrier support **502** and second side carrier support **504**. The first piece ends at the junction, and the remaining portion of first side carrier support **502** and second side carrier support **504** in combination with the second vertical supports **508**, **512** and second base element **516** may be referred to as the second piece of the car seat frame **500**. These elements may be constructed as illustrated, or by any other arrangement. They may all be integral with one another, or divided into two or separate pieces that are fixed together by any suitable fastener or junction element. The material of these elements may be any material sufficient to create the structure, and withstand the anticipated weight of a pet carrier and pet set on the car seat frame. For example, the material may be metal or polymeric. The material may be hollow or solid. The material may be hollow tubing.

A car seat frame may comprise one or more seat belt retainers. Seat belt retainers **522a** and **522b** are illustrated in FIG. 5 and may be used to secure the car seat frame **500** on a car seat with the seat belt fitted with the retainers. Seat belt retainers **524a**, **524b** are optional for the FIG. 5 embodiment, and if present would allow securing the car seat frame **500** to the seat in the opposite direction. Seat belt retainers may be integral with or separate from the vertical supports.

A car set may be secured to a car seat frame in embodiments herein in many different fashions. For example, a car seat could be strapped to a car seat frame. The car seat frame **500** comprises a first frame-carrier receiver **518** fixed to the first side carrier support **502** and a second frame-carrier receiver **520** fixed to the second side carrier support **504**. The frame-carrier receivers may engage matching engagements of a carrier. For example, the frame-carrier receiver **518** and frame-carrier receiver **520** may engage pet carrier engagement **120a** and pet carrier engagement **120b** of pet carrier **100** (see FIGS. 1-4).

An embodiment includes an assembly comprising a pet carrier and a car seat frame, where the pet carrier is secured to the car seat frame. Optionally, the car seat frame is also secured to a seat. The seat may be the seat of a car. The car seat frame could be secured to the seat of the car by wrapping a seat belt of the seat around elements of the frame. An embodiment of the assembly includes the pet carrier **100** secured to the car seat frame **500**, as illustrated in FIG. 9. As illustrated in FIG. 9, the frame-carrier receiver **518** and frame-carrier receiver **520** engage pet carrier engagement **120a** and pet carrier engagement **120b** of pet carrier **100**. The assembly of pet carrier **100** and car seat frame **500** may further comprise a seat comprising a seat belt, where the seat belt is arranged in the seat belt retainer **522a** and seat belt retainer **522b**, or the optional seat belt retainer **524a** and seat belt retainer **524b**, to secure the car seat frame **500** to the seat. The seat may be a car seat.

An embodiment includes a pet stroller that may carry a pet carrier. The pet stroller may include wheels in any configuration and supports on which a pet carrier may rest. It may also include restraints to secure the pet carrier to the pet stroller. Referring to FIGS. 6-8, an embodiment is illustrated as pet stroller **600**, which comprises a frame **601** and four wheels (wheel **658**, wheel **660**, wheel **662**, and wheel **664**).

As illustrated, the frame comprising a first side **602**, a second side **604**, a push bar **631**, a front pet carrier support **624**, and a rear pet carrier support **628**. Each of the first side **602** and the second side **604** comprises a front lower portion (**610**, **616**), a rear lower portion (**611**, **620**), a mid-portion (**614**, **615**), and an upper portion (**612**, **613**). Each of the front and rear lower portions have a first end and a wheel engagement at the first end to which one of the wheels is engaged. The front **610** and rear **611** lower portions on the first side each having a second end opposite the respective first ends that converge at a first junction **621a**. The front **616** and rear **620** lower portions on the second side each have a second end opposite the respective first end that converges at a second junction **621b**. The mid portion **614** on the first side has a first end at the first junction **621a** and a second end extended in a direction generally opposite to the direction in which the lower portion on the first side extends from the first junction. The mid portion **615** on the second side has a first end at the second junction **621b** and a second end extending in a direction generally opposite to the direction in which the lower portion on the second side extends from the second junction. The upper portion **612** on the first side is illustrated with a first end extending from the second end of the mid portion **614** on the first side and a second end, the

upper portion **613** on the second side having a first end extending from the second end of the mid portion **615** on the second side and a second end. The push bar **631** is illustrated with a first end connected to the second end of the upper portion **612** on the first side, and a second end connected to the second end of the upper portion **613** on the second side. The specific arrangement illustrated may be varied in embodiments herein. For example, other angles between the above mention elements are within the scope of embodiments herein.

The frame may be made of separate elements fixed to one another. Alternatively, one or more of the elements may be integral with one another. The material of the frame may be any that supports the utility of the pet stroller. The frame may be comprised completely or in part by elements that are polymeric or metal. The frame may be comprised completely or in part by elements that are either hollow or solid. Any suitable fasteners may be used to fasten separate frame elements to one another. For example, fastener may be chosen from at least one of rivets, nuts and bolts, screws, junction elements made to fit and secure the various frame elements, etc. This type of material and construction may apply to all other elements of the pet stroller **800**. For example, the front pet carrier support **624**, the rear pet carrier support **628**, the first intermediate cross member **626**, and rear cross member **810** may be fashioned in a similar manner. Connectors **634**, **636**, **638**, **640**, **642**, **644**, **646**, and **648** are illustrated in FIGS. 6-8. A connector may connect separate pieces of a pet stroller. Alternatively, a connector may be positioned on a single element to provide attachment points for other elements.

The front pet carrier support **624** is illustrated extending between and fixed to the front lower portions **610**, **616**. Similarly, the rear pet carrier support **628** is illustrated extending between and fixed to the rear lower portions **611**, **620**.

A pet stroller may simply provide a space for a pet carrier, and optionally supports for a pet stroller. A user then could chose to secure the pet carrier to the pet stroller in any suitable fashion. For example, a pet carrier could be strapped to elements of the pet stroller. The pet stroller embodiment of FIGS. 6-8 comprise first stroller-carrier receiver **674** fixed to the first junction **621a** and facing the second junction **621b**. The pet stroller of FIGS. 6-8 also comprises a second stroller-carrier receiver **676** fixed to the second junction **621b** and facing the first junction **621a**. The first stroller-carrier receiver **674** and the second stroller-carrier receiver are capable of engaging pet carrier engagements in order to fix a pet carrier to the pet stroller.

An embodiment includes an assembly comprising a pet carrier and a pet stroller, where the pet carrier is received in or secured to the pet stroller. The pet carrier may be secured to the pet stroller by any suitable securements. For example, the pet carrier could be lashed to elements of the pet stroller. An embodiment of the assembly comprises the pet carrier **100** secured to the pet stroller **600**. See FIG. 10. As illustrated in FIG. 10, the first and second pet carrier engagements **120a**, **120b** are respectively engaged with the first and second stroller-carrier receivers **674**, **676**.

An embodiment includes a system comprising a pet carrier, a car seat frame, and a pet stroller. The car seat frame and the pet stroller may be capable of receiving the pet carrier for transport of the pet carrier on either a car seat or the pet stroller. Any suitable method of receiving and/or securing the pet carrier to either the car seat frame or the pet stroller may be part of the system. The pet carrier may comprise one or more engagements that match one or more

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receivers on one or both of the car seat carrier and the pet stroller. The pet carrier may include one engagement type that matches receiver(s) on the car seat carrier, and a second engagement type that matches receiver(s) on the pet stroller. The engagement(s) of the pet carrier may be of one type, and the receivers on both the car seat carrier and the pet stroller may be configured to match the one type of engagement(s) on the pet carrier. The system may comprise the pet carrier 100, the car seat frame 500, and the pet stroller 600. As illustrated, the first stroller-carrier receiver 674, the second stroller-carrier receiver 676, the first frame-carrier receiver 518, and the second frame-carrier receiver 520 are illustrated as matching such that the pet carrier 100 may be assembled with either the car seat frame 500 or the pet stroller 600.

As stated above, the carrier engagements 120a, 120b may have alternate structures/mechanisms and/or be moved forward or backward along the pet carrier so that the pet carrier can fit to various secondary devices. The carrier engagements herein are non-limiting examples of carrier engagements or carrier engagement means of embodiments herein. Likewise, the carrier receivers and carrier supports on secondary devices may have alternate structures/mechanisms and/or be moved forward or backward along their respective elements to accommodate alternate carrier engagements. The receivers and carrier supports herein are non-limiting examples of receivers or receiver means and carriers or carrier means of embodiments herein. Pet carriers, car seat frames, and strollers may have alternate accessory structures/mechanisms as well. Non-limiting examples include a car seat frame that may have alternate seat belt retainers, a car seat frame that may have an alternate configuration of supports positioning the frame-carrier receivers at the correct height, or a stroller with front and back carrier supports contiguous with one another to encircle a carrier.

Referring to FIG. 11, an embodiment of a car seat frame 1101 exemplifying alternate frame-carrier receivers and alternate seat belt retainers is illustrated. A frame-carrier receiver 1118, on a first side carrier support 1102, is illustrated opposed to a frame-carrier receiver 1120, on a second side carrier support 1104. The frame-carrier receiver comprises a retainer 1118a and a retainer 1118b. In the embodiment illustrated, the retainer 1118a includes an inclined surface 1119a, and the retainer 1118b includes an inclined surface 1119b. The two inclined surfaces 1119a and 1119b are angled toward each other. In this configuration, a pet carrier engagement of a carrier being positioned upon the car seat frame may be guided into the frame carrier receiver 1118. The retainers may also engage with structures on a pet carrier to aid in position and/or retention of the pet carrier. A frame carrier receiver 1120 is illustrated opposed to the frame-carrier receiver 1118. Like the frame-carrier receiver 1118, the frame-carrier receiver 1120 includes retainers 1120a and 1120b, with inclined surfaces 1121a and 1121b.

FIGS. 11 and 14 also illustrate alternate seat belt retainers. One or more of seat belt retainers 1122a, 1122b, 1124a, and 1124b may be included on a car seat frame herein. Referring to seat belt retainer 1122a, the retainer may include an elongated body comprising a slot 1401 and a slot 1403. The retainer may also include a protrusion 1405 and a protrusion 1407 on a vertical support of the car seat frame. The protrusion 1405 and the protrusion 1407 may be spaced such that a seat belt may be arranged between them. The protrusion 1405 and the protrusion 1407 may also be spaced such that the slot 1401 and the slot 1403 of the elongated body may be respectively fitted thereon. As illustrated, the slots 1401, 1401 may be configured to slide from a widened opening that receives a larger end of the protrusion 1405,

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1407, respectively, down to a narrowed opening over a narrowed portion of the protrusion 1405, 1407, thereby locking the elongated body onto the protrusions. A seat belt may be aligned between the protrusion 1405 and 1407 prior to engaging the elongated body, and in such a manner lock the seat belt to the car seat frame. The seat belt retainer 1122a also includes a user control tab 1409, which may be engaged by a user to operate the seat belt retainer 1122a. One seat belt retainer is illustrated in FIG. 14. One or more seat belt retainers of a car seat frame may be similarly configured.

FIG. 12 illustrates a pet carrier 1299 with a pet carrier engagement 1201 in combination with the car seat frame of FIG. 11. As illustrated, the pet carrier engagement 1201 is positioned between the retainers 1118a and 1118b.

FIG. 13 illustrates the pet carrier engagement 1201 on the pet carrier illustrated in FIG. 12 engaged with the frame-carrier receiver, like that of frame-carrier receiver 1120 in FIG. 11. The pet carrier engagement 1201 includes a mount 1301, which is fixed to the bottom section 1304 of a pet carrier 1299. A mount may, as illustrated for mount 1301, be fixed to the a first bottom frame element 1314. A pet carrier engagement may also include, as illustrated for pet carrier engagement 1201, a rotating latch 1380. The rotating latch may be connected to a mount. In the embodiment illustrated, the rotating latch 1380 includes pins received in holes within the mount 1301. One pin 1381 is illustrated in one hole 1382. A similar pin and hole to match is on the far side of the rotating latch 1380/mount 1301 combination (not shown). The rotating latch 1380 includes an opening (see FIG. 16) that may be fitted around the second side carrier support 1104. The open position of rotating latch 1380 in FIG. 16 is shown in relation to a stroller. However, a similar configuration may be envisioned between the rotating latch 1380 and the second side carrier support 1104. Once the rotating latch is in position over the support it may be rotated to close upon the support. In FIG. 13, the rotating latch 1380 is illustrated in a position where it has been fitted around the second side carrier support 1104. It may be rotated further downward to a closed position. It may lock in the closed position. The lock may be any suitable lock. The lock may be press fitted engagements that may be released by flexing the rotating latch 1380 or forcing the engagements past one another. A pet carrier may include a pet carrier engagement like pet carrier engagement 1201 on each side, in place of pet carrier engagements 120a and 120b of FIGS. 1 and 2.

FIGS. 15-20 illustrate the pet carrier 1299 having alternate pet carrier engagements and a stroller 1501. The pet carrier engagement 1201 is illustrated on one side of the pet carrier 1299, and a pet carrier engagement 1801 is illustrated on the other side of the pet carrier engagement 1201.

Referring to FIG. 17, the stroller 1501 includes a carrier support 1503. The carrier support 1503 may include a carrier support frame 1505 including a first side support 1510 connected to the first side of the stroller and having a front end 1511 and a rear end 1512, and a second side support 1520 connected to the second side of the stroller and having a front end 1521 and rear end 1522. The carrier support may also include a third side 1530 extending from the front end 1511 of the first side support 1510 and the front end 1521 of the second side support 1520, and a fourth side 1540 extending from the rear end 1512 of the first side support 1510 and the rear end 1522 of the second side support 1520. A stroller herein may be configured as known to the skilled artisan to present a wheeled and pushable support system for a pet carrier. The basic elements may be similar to those of the stroller of FIG. 6. Alternatively, a carrier support 1503,

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as illustrated in FIGS. 15-20, may replace the front carrier support 624 and the rear carrier support 628. The two different carrier support configurations for the strollers of FIG. 6 versus FIGS. 15-20 represent non-limiting means by which a stroller may support a carrier. Generally, a stroller may comprise a carrier support. The carrier support may be two or more different structures. See the non-limiting example of FIG. 6. The carrier support may be a single support (which may comprise a single integral unit, or be comprise of parts engaged together to form the single support). See the non-limiting example of FIGS. 15-20. The support(s) represent non-limiting examples of a support(s) or support means of embodiments herein.

FIG. 15 illustrates the pet carrier 1299 in combination with the stroller 1501. The pet carrier includes the pet carrier engagement 1201, but in FIG. 15 it is engaged on the carrier support 1503 of a stroller 1501. The engagement of the carrier support 1503 by the pet carrier engagement 1201 is similar to that described above for the car set frame in FIGS. 11-13. FIG. 15 illustrates the pet carrier engagement 1201 closed upon the carrier support 1503. FIG. 16 illustrates the pet carrier engagement 1201 fitted onto the carrier support 1503 but in the open position.

FIGS. 17-20 illustrate different views of the stroller 1501 and the pet carrier 1299. The carrier support 1503 may prevent forward or reward movement of the pet carrier 1299. The carrier support may include first and second stroller-carrier receivers. The first and second stroller-carrier receivers may be, but are not necessarily limited to ones similar or the same in configuration and operation as the frame carrier receivers 1118 and 1120, or the stroller-carrier receivers 674, 676.

As stated above, an embodiment includes an assembly comprising a pet carrier and a pet stroller, where the pet carrier is received in or secured to the pet stroller. The pet carrier may be secured to the pet stroller by any suitable securements. For example, the pet carrier could be lashed to elements of the pet stroller. An embodiment of the assembly comprises the pet carrier 1299 secured to the pet stroller 1501.

An embodiment includes an assembly comprising a pet carrier and a car seat frame, where the pet carrier is received in or secured to the car seat frame. The pet carrier may be secured to the car seat frame by any suitable securements. For example, the pet carrier could be lashed to elements of the car seat frame. An embodiment of the assembly comprises the pet carrier 1299 secured to the car seat frame 500. An embodiment of the assembly comprises the pet carrier 100 secured to the car seat frame 1101.

An embodiment includes a system comprising a pet carrier, a car seat frame, and a pet stroller. The car seat frame and the pet stroller may be capable of receiving the pet carrier for transport of the pet carrier on either a car seat or the pet stroller. Any suitable method of receiving and/or securing the pet carrier to either the car seat frame or the pet stroller may be part of the system. The pet carrier may comprise one or more engagements that match one or more receivers on one or both of the car seat carrier and the pet stroller. The pet carrier may include one engagement type that matches receiver(s) on the car seat carrier, and a second engagement type that matches receiver(s) on the pet stroller. The engagement(s) of the pet carrier may be of one type, and the receivers on both the car seat carrier and the pet stroller may be configured to match the one type of engagement(s) on the pet carrier. The system may comprise the pet carrier 100, the car seat frame 500, and the pet stroller 600. As illustrated, the first stroller-carrier receiver 674, the second

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stroller-carrier receiver 676, the first frame-carrier receiver 518, and the second frame-carrier receiver 520 are illustrated as matching such that the pet carrier 100 may be assembled with either the car seat frame 500 or the pet stroller 600. The system may comprise the pet carrier 1299, the car seat frame 1101, and the pet stroller 1501. As illustrated, the pet carrier engagements 1201 and 1801 can be used to secure the pet carrier 1299 to either the car seat frame 1101 or the pet stroller 1501.

The following list of embodiments is a non-exclusive gathering of embodiments herein. Embodiments otherwise described herein are not excluded from the scope of what may be claimed by this list.

LIST OF EMBODIMENTS

1. A pet carrier comprising a top section, a bottom section having an upper rim, a handle, a first pet carrier engagement, a second pet carrier engagement, and an interior defined between the top section and the bottom section; the top section comprising a front, a back, a canopy, at least one canopy frame element supporting the canopy, a front canopy latch element at the front, and a back canopy latch element at the back, the canopy comprising a lower edge configured to be proximal to the upper rim of the bottom section, a first canopy frame element of the at least one canopy frame element extending at least partially around the lower edge of the canopy and having a front section at the front and a back section at the back, the front section and the back section meeting at a first pivot on a first side of the canopy and meeting at a second pivot on a second side of the canopy, the front canopy latch element fixed to the front section and the back canopy latch element fixed to the back section; the bottom section comprising a base, a front, a back, a first side, a second side, a shell, a bottom frame element, a front bottom latch element on the front, and a back bottom latch element on the back, the shell extends up from the base to the bottom frame element, the bottom frame element extends around the upper rim of the bottom, and the front and the back bottom latch elements are fixed to the bottom frame element; the handle comprising a first handle receiver engaged with one or more of the first canopy frame element and the first bottom frame element on a first side of the pet carrier, and a second handle receiver engaged with one or more of the first canopy frame element and the first bottom frame element on a second side of the pet carrier, a handle bar extending from the first handle receiver to the second handle receiver and over the canopy; the front canopy latch element being engageable with the front bottom latch element to secure the front of the top to the front of the bottom in a front closed position, and being releasable to an open position, allowing movement of the top section at the front of the pet carrier to reveal the interior of the pet carrier; the back canopy latch element being engageable with the back bottom latch element to secure the back of the top to the back of the bottom in a back closed position, and being releasable to an open position, allowing movement of the top section at the back of the pet carrier to reveal the interior of the pet carrier; the first pet carrier engagement on the first side of the bottom section, the second pet carrier engagement on the second side of the bottom section, both the first pet carrier engagement and the second pet carrier engagement being engageable with a secondary device in order to fix the pet carrier to the secondary device; and the top section, the bottom section, and the base forming an enclosure, in which

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a pet may be contained, within the interior of the pet carrier when the front and the back of the canopy are in the closed position.

2. The pet carrier of embodiment 1 further comprising a first support wire proximal to the front of the bottom section, engaged with the first bottom frame element on first and second sides of the bottom section, and extending from the first bottom frame element along the first side of the bottom toward the base, across the base, and along the second side of the bottom toward the first bottom frame element, and a second support wire proximal to a rear of the bottom section, engaged with the first bottom frame element on the first and second sides of the bottom, and extending from the first bottom frame element along the first side of the bottom toward the base, across the base, and along the second side of the bottom toward the first bottom frame element, one or both of the first support wire and second support wire may be positioned either exterior to the bottom or interior to the bottom.

3. The pet carrier of embodiment 1 or embodiment 2, wherein the first pet carrier engagement extends toward the base, and the second pet carrier engagement extends toward the base.

4. The pet carrier of embodiment 1 or embodiment 2, wherein the first pet carrier engagement comprises a first mount fixed to the bottom section, and a first rotating latch rotatably fixed to the first mount and configured to engage the secondary device, and the second pet carrier engagement comprises a second mount fixed to the bottom section, and a second rotating latch rotatably fixed to the second mount and configured to engage the secondary device.

5. A car seat frame comprising a first side, a second side, a first base element having a first end and a second end, and a second base element having a first end and a second end; the first side comprising a first side carrier support, a first vertical support on the first side, and a second vertical support on the first side, each vertical support on the first side having a first end and a second end, the first end of each fixed to opposite ends of the first side carrier support; the second side comprising a second side carrier support, a first vertical support on the second side, and a second vertical support on the second side, each vertical support on the second side having a first end and a second end, the first end of each fixed to opposite ends of the second side carrier support; and the second end of the first vertical support on the first side is fixed to the first end of the first base element, the second end of the second vertical support on the first side is fixed to the first end of the second base element, the second end of the first vertical support on the second side is fixed to the second end of the first base element, and the second end of the second vertical support on the second side is fixed to the second end of the second base element; the car seat frame further comprising a first frame-carrier receiver fixed to the first side carrier support and a second car seat frame-carrier receiver fixed to the second side carrier support.

6. The car seat frame of embodiment 5, wherein the first frame-carrier receiver comprises a first retainer and a second retainer fixed on the first side carrier support, and a first side carrier support length of the first carrier support positioned between the first retainer and the second retainer, the first retainer and the second retainer spaced to receive a first pet carrier engagement therebetween, the second frame-carrier receiver comprises a third retainer and a fourth retainer fixed on the second side carrier support, and a second side carrier support length of the second side carrier support positioned between the third retainer and the fourth retainer, the third

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retainer and the fourth retainer spaced to receive a second pet carrier engagement therebetween.

7. The car seat frame embodiment 5 or embodiment 6 comprising one or more a seat belt retainers, where each seat belt retainer is fixed to one of the vertical supports.

8. The car seat frame of embodiment 7, wherein the one or more seat belt retainers comprise a first seat belt retainer fixed to the first vertical support on the first side and a second seat belt retainer fixed to the second vertical support on the first side.

9. An assembly comprising a pet carrier and the car seat frame of any one of embodiments 5-8.

10. The assembly of embodiment 9, wherein the pet carrier is the pet carrier of one of embodiments 1-4.

11. The assembly of embodiment 10, wherein the first pet carrier engagement is engaged with the first frame-carrier receiver, and the second pet carrier engagement is engaged with the second frame-carrier receiver.

12. A pet stroller comprising a frame and four wheels; the frame comprising a first side, a second side, a push bar, a carrier support, each side comprising a front lower portion, a rear lower portion, a mid portion, and an upper portion; each of the front and rear lower portions having first end and a wheel engagement at the first end to which one of the wheels is engaged; the front and rear lower portions on the first side each having a second end opposite the respective first ends that converge at a first junction, the front and rear lower portions on the second side each having a second end opposite the respective first end that converges at a second junction; the mid portion on the first side having a first end at the first junction and a second end extended in a direction generally opposite to the direction in which the lower portion on the first side extends from the first junction, the mid portion on the second side having a first end at the second junction and a second end extending in a direction generally opposite to the direction in which the lower portion on the second side extends from the second junction; the upper portion on the first side having a first end extending from the second end of the mid portion on the first side and a second end, the upper portion on the second side having a first end extending from the second end of the mid portion on the second side and a second end; the push bar having a first end connected to the second end of the upper portion on the first side, and a second end connected to the second end of the upper portion on the second side; wherein the carrier support is configured to receive a pet carrier.

13. The pet stroller of embodiment 12, wherein the carrier support comprises a frame defining a rectangular opening into which a pet carrier may be received, a first side of the frame connected to the first side of the stroller and a second side of the frame connected to the second side of the stroller, and wherein the pet carrier engagements of a pet carrier may be received by the carrier support.

14. The pet stroller of embodiment 13, wherein the first side of the frame is connected to the mid portion on the first side of the stroller and a second side of the frame is connected to the mid portion on the second side of the stroller.

15. The pet stroller of embodiment 12, wherein the pet stroller further comprises a first stroller-carrier receiver fixed to the first junction and facing the second junction, and a second stroller-carrier receiver fixed to the second junction and facing the first junction, the first and second stroller-carrier receivers capable of engaging carrier engagements in order to fix a pet carrier to the pet stroller.

16. The pet stroller of embodiment 12 or embodiment 15, wherein the carrier support comprises a front pet carrier

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support extending between and fixed to the front lower portions, and a rear pet carrier support extending between and fixed to the rear lower portions.

17. An assembly comprising a pet carrier and the pet stroller of one of embodiments 12-16, wherein the pet carrier 5 engagements engage the pet stroller.

18. The assembly of embodiment 17, wherein the pet carrier is the pet carrier of one of embodiments 1-4.

19. The assembly of embodiment 18, wherein the first and second pet carrier engagements are respectively engaged 10 with either the carrier support or the first and second stroller-carrier receivers.

20. A system comprising at least two of the pet carrier of one of embodiments 1-4, the car seat frame of one or more of embodiments 5-8, the assembly of one or more of 15 embodiments 9-11, the pet stroller of one of embodiments 12-16, and the assembly of one of embodiments 17-19.

Further embodiments may be achieved by changing a structure in an embodiment herein to a corresponding structure in another embodiments herein. For example, the pet 20 carrier engagements of FIGS. 1-3 may be used in systems herein in combination with car seats or strollers with receivers like FIGS. 4-10 but fitted onto a support like that of support 1503 of FIGS. 15-20.

REFERENCE CHARACTER LISTING

100—Pet Carrier
 102—Top Section
 104—Bottom Section
 105—Shell
 106—Handle
 107a—First Handle Receiver
 107b—Second Handle Receiver
 107c—First Handle Release
 107d—Second Handle Release
 108—Canopy
 109—Lower Edge
 110a—Front Canopy Latch Element
 110b—Rear Canopy Latch Element
 111—First Canopy Frame Element
 111a—Ridge
 111b—Ridge
 111c—Ridge
 111d—Ridge
 112—Carrier Grip
 114—First Bottom Frame Element
 115a—Front Bottom Latch Element
 115b—Back Bottom Latch Element
 116—First Support Wire
 118—Second Support Wire
 120a—Pet Carrier Engagement
 120b—Pet Carrier Engagement
 124—Base
 125—Upper Rim
 126—Front Section of First Canopy Frame Element
 127—Rear Section of First Canopy Frame Element
 128—First Pivot
 129—Second Pivot
 130—Handle Bar
 131—Grip
 210a—Wire Holder
 210b—Wire Holder
 310a—Wire Holder
 310b—Wire Holder
 500—Car Seat Frame
 502—First Side Carrier Support

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504—Second Side Carrier Support
 506—First Vertical Support on the First Side
 508—Second Vertical Support on the First Side
 510—First Vertical Support on the Second Side
 512—Second Vertical Support on the Second Side
 514—First Base Element
 516—Second Base Element
 518—Frame-Carrier Receiver
 520—Frame-Carrier Receiver
 522a—Seat Belt Retainer
 522b—Seat Belt Retainer
 524a—Seat Belt Retainer
 524b—Seat Belt Retainer
 600—Pet stroller
 601—Frame
 602—First Side
 604—Second Side
 610—Front First Side Lower Portion
 611—Rear First Side Lower Portion
 612—Upper Portion
 613—Upper Portion
 614—Mid Portion
 615—Mid Portion
 616—Front Second Side Lower Portion
 620—Rear Second Side Lower Portion
 621a—First Junction
 621b—Second Junction
 622—Brace
 624—Front Pet Carrier Support
 625—Brace
 626—First Intermediate Cross Member
 628—Rear Pet Carrier Support
 630—Grip
 631—Push Bar
 632—Release
 634—Connector
 636—Connector
 638—Connector
 640—Connector
 642—Connector
 644—Connector
 646—Connector
 648—Connector
 650—Front Panel
 652—First Side Panel
 654—Second Side Panel
 656—Bottom Panel
 658—Wheel
 660—Wheel
 662—Wheel
 664—Wheel
 666—Wheel Housing
 668—Wheel Housing
 670—Front Wheel Bearing Support
 672—Front Wheel Bearing Support
 674—First Stroller-Carrier Receiver
 676—Second Stroller-Carrier Receiver
 810—Rear Cross Member
 1101—Car Seat Frame
 1102—First Side Carrier Support
 1104—Second Side Carrier Support
 1118—Frame-Carrier Receiver
 1118a—Retainer
 1118b—Retainer
 1119a—Inclined Surface
 1119b—Inclined Surface
 1120—Frame-Carrier Receiver

1120a—Retainer
 1120b—Retainer
 1121a—Inclined Surface
 1121b—Inclined Surface
 1201—Pet Carrier Engagement 1201
 1299—Pet Carrier
 1301—Mount
 1304—Bottom Section
 1314—First Bottom Frame Element
 1380—Rotating Latch
 1501—Stroller
 1503—Carrier Support
 1801—Pet Carrier Engagement

It is understood, therefore, that the invention is not limited to the particular embodiments disclosed, but is intended to cover all modifications which are within the spirit and scope of the invention as defined by the appended claims, the above description, and/or shown in the attached drawings.

The invention claimed is:

1. An assembly comprising a pet stroller with a removably attached pet carrier:

the pet stroller comprising:

a stroller frame comprising a first side, a second side, a push bar, a carrier support, each side comprising a front lower portion, a rear lower portion, a mid portion, and an upper portion;

each of the front and rear lower portions having a first end and a wheel engagement at the first end to which a wheel is engaged;

the front and rear lower portions on the first side each having a second end opposite the respective first ends that converge toward one another, the front and rear lower portions on the second side each having a second end opposite the respective first end that converge toward one another; and

the pet carrier comprising:

a top section, a bottom section having an upper rim, a first pet carrier engagement, a second pet carrier engagement, and an interior defined between the top section and the bottom section;

the top section comprising a front, a back, a canopy, at least one canopy frame element supporting the canopy, the canopy comprising a lower edge configured to be proximal to the upper rim of the bottom section, a first canopy frame element of the at least one canopy frame element extending at least partially around the lower edge of the canopy and having a front section at the front and a pivotal connection to the upper rim;

the bottom section comprising a base, a front, a back, a first side, a second side, a shell, and a bottom frame element, the shell extends up from the base to the bottom frame element, the bottom frame element extends around and defines the upper rim of the bottom;

at least one of the top section at the front of the pet carrier or the top section at the back of the pet carrier is openable to reveal the interior of the pet carrier;

the first pet carrier engagement at the first side of the bottom section, the second pet carrier engagement at the second side of the bottom section, both the first pet carrier engagement and the second pet carrier engagement being engageable with the carrier support of the stroller; and

the top section, the bottom section, and the base forming an enclosure, in which a pet may be contained

within the interior of the pet carrier when the front and the back of the canopy are in a closed position; wherein the first pet carrier engagement comprises a first mount fixed to the bottom section, the first mount including hooked protrusions extending at either end of the first mount and a first rotating latch rotatably coupled to the first mount between the hooked protrusions, the hooked protrusions of the first mount and the first rotating latch are configured to engage a first side support of the carrier support of the stroller.

2. The assembly of claim 1, wherein the second pet carrier engagement comprises a second mount fixed to the bottom section, the second mount including hooked protrusions extending at either end of the second mount and a second rotating latch rotatably coupled to the second mount between the hooked protrusions, the hooked protrusions of the second mount and the second rotating latch are configured to engage a second side support of the carrier support of the stroller.

3. An assembly comprising a pet stroller with a removably attached pet carrier:

the pet stroller comprising:

a stroller frame comprising a first side, a second side, a push bar, a carrier support, each side comprising a front lower portion, a rear lower portion, a mid portion, and an upper portion;

each of the front and rear lower portions having a first end and a wheel engagement at the first end to which a wheel is engaged;

the front and rear lower portions on the first side each having a second end opposite the respective first ends that converge toward one another, the front and rear lower portions on the second side each having a second end opposite the respective first end that converge toward one another;

each mid portion being a singular bar having a first end connected to the second end of the respective front lower portion and a second end that extends in an opposite direction;

the upper portion on the first side having a first end extending from the second end of the mid portion on the first side and a second end, the upper portion on the second side having a first end extending from the second end of the mid portion on the second side and a second end;

the push bar having a first end connected to the second end of the upper portion on the first side, and a second end connected to the second end of the upper portion on the second side; and

a first brace extending from the rear lower portion on the first side to the upper portion on the first side and a second brace extending from the rear lower portion on the second side to the upper portion on the second side wherein

the carrier support is located in a region of the mid portions and comprises a carrier support frame including a first side support connected only directly to the first side mid portion and the first brace and having a front end and a rear end, and a second side support connected only directly to the second side mid portion and the second brace and having a front end and rear end, a third side extending from the front end of the first side support to the front end of the second side support, and a fourth side extending from the rear end of the first side support to the rear end of the second side support,

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the carrier support configured to receive the pet carrier within the carrier support frame; and
 the pet carrier comprising:
 a top section, a bottom section having an upper rim, a first pet carrier engagement, a second pet carrier engagement, and an interior defined between the top section and the bottom section;
 the top section comprising a front, a back, a canopy, at least one canopy frame element supporting the canopy, the canopy comprising a lower edge configured to be proximal to the upper rim of the bottom section, a first canopy frame element of the at least one canopy frame element extending at least partially around the lower edge of the canopy and having a front section at the front and a pivotal connection to the upper rim;
 the bottom section comprising a base, a front, a back, a first side, a second side, a shell, and a bottom frame element, the shell extends up from the base to the bottom frame element, the bottom frame element extends around and defines the upper rim of the bottom;
 at least one of the top section at the front of the pet carrier or the top section at the back of the pet carrier is openable to reveal the interior of the pet carrier;
 the first pet carrier engagement at the first side of the bottom section, the second pet carrier engagement at the second side of the bottom section, both the first pet carrier engagement and the second pet carrier engagement being engageable with the carrier support of the stroller; and
 the top section, the bottom section, and the base forming an enclosure, in which a pet may be contained within the interior of the pet carrier when the front and the back of the canopy are in the closed position;
 wherein the bottom section further comprises a front canopy latch on the front, the front canopy latch protruding outwardly from the bottom frame element, and the front canopy latch having a shape of a rectangular cuboid;
 wherein the first pet carrier engagement comprises a first mount fixed to the bottom section, the first mount including hooked protrusions extending at either end of the first mount and a first rotating latch rotatably coupled to the first mount between the hooked protrusions, the hooked protrusions of the first mount and the first rotating latch are configured to engage the first side support of the carrier support of the stroller.

4. The assembly of claim 3, wherein the front canopy latch is engageable to secure the front of the top to the front

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of the bottom in a front closed position, and being releasable to an open position, allowing movement of the top section at the front of the pet carrier to reveal the interior of the pet carrier.

5. The assembly of claim 4, wherein the back of the top is connectable to the back of the bottom in a back closed position, and is releasable to an open position, allowing movement of the top section at the back of the pet carrier to reveal the interior of the pet carrier.

6. The assembly of claim 3, wherein the first and second pet carrier engagements are engaged with the carrier support.

7. The assembly of claim 6 further comprising a lower carrier support on the pet stroller, the lower carrier support comprising a front frame connected to the front lower portions, and a rear frame connected to the rear lower portions.

8. The assembly of claim 3 further comprising a handle connected to the pet carrier.

9. The assembly of claim 7, wherein the lower carrier support supports the bottom frame element of the pet carrier.

10. The assembly of claim 7 further comprising a storage area located beneath the lower carrier support, the storage area including a flexible bottom and two side panels, one located on each of the first and second sides of the stroller frame.

11. The assembly of claim 3, wherein the carrier support frame includes a continuous perimeter.

12. The assembly of claim 3, wherein the carrier support frame is oriented in a singular flat plane.

13. The assembly of claim 3, wherein the second pet carrier engagement comprises a second mount fixed to the bottom section, the second mount including hooked protrusions extending at either end of the second mount and a second rotating latch rotatably coupled to the second mount between the hooked protrusions, the hooked protrusions of the second mount and the second rotating latch are configured to engage the second side support of the carrier support of the stroller.

14. The assembly of claim 3, wherein the rectangular cuboid shape of the front canopy latch includes rounded corners.

15. The assembly of claim 3, wherein the top section further comprises a top front canopy latch on the front, the top front canopy latch protruding outwardly from the first canopy frame element, and the top front canopy latch having a shape of a rectangular cuboid.

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